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# A novel stochastic neural network framework for modeling and simulation of within-host Chikungunya virus transmission with latency

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ABSTRACT

In this study, we present an innovative stochastic computing framework for modeling and simulation of within-host transmission dynamics of Chikungunya virus infection, incorporating latency and randomness. We integrate a feedforward neural network with the Legendre spectral collocation method to provide accurate results for the complex nonlinear model. The stochastic model captures intrinsic randomness associated with disease progression. A rigorous theoretical analysis is conducted, establishing the stability of the disease-free equilibrium when the control reproduction number $\tilde{R}_0 < 1$. To explore the model's behavior under stochastic influences, we implement the spectral collocation scheme for numerical simulations and analyze the impact of key epidemiological parameters. Extensive computational experiments are performed to support the theoretical results. The effectiveness and reliability of the present neural network-enhanced scheme are assessed through multiple evaluation criteria, including regression performance, mean squared error (MSE), error distribution, and phase portrait analysis. Additionally, the developed approach is benchmarked against the standard spectral collocation method to demonstrate its improved accuracy and predictive capabilities. The present scheme can be effectively extended to simulate complex real-world systems beyond epidemic models.

## 1. Introduction

Chikungunya is a viral infectious disease caused by the Chikungunya virus (CHIKV), an RNA virus belonging to the Togaviridae family. This infection is primarily transmitted through the bites of infectious Aedes mosquitoes, particularly Aedes aegypti and Aedes albopictus, which are also vectors for other arboviral diseases including dengue and Zika [1]. This viral infection was first reported in 1952 in Tanzania and has since triggered multiple outbreaks across Africa and Asia [2,3]. The recent emergence of this infection pose a significant health threat in many countries worldwide. A major outbreak in India led to approximately 1.3 million confirmed cases during 2005 to 2006 [4]. In 2011, the outbreak of CHIKV had expanded to Southern and Southeastern Asia, Central Africa, the Pacific region, Europe, and the islands of the Western Indian Ocean. The symptoms of this viral disease are closely resemble those of Dengue and Zika viral infections, often leading to misdiagnosis the CHIKV. The infection typically causes fever along with a severe joint pain, which can be debilitating and vary in duration. Due to challenges in diagnosis and reporting, the true burden of the disease is likely under estimated. While severe symptoms and fatalities are rare, they are usually associated with underlying health conditions [5]. Despite a significant raise in incidence and morbidity associated with CHIKV infection, no specific antiviral treatment or commercially available vaccine has been developed in the scientific community till date [6]. However, several vaccine candidates are in different stages of preclinical and clinical development [7].

Mathematical modeling plays an important role in analyzing real-life phenomena, including control of infectious diseases [8]. These models are typically formulated using differential systems based on integer-order [9,10] or fractional-order modeling approaches [11–13]. Numerous mathematical models have been studied and rigorously analyzed on analyzing and controlling CHIKV infection. The population-level dynamics and control of CHIKV, considering viral mutation, were studied in [14]. Optimal control analysis applied to vector-borne infectious diseases, with a case study of CHIKV in Chad, was presented in [15]. The dynamics of CHIKV incorporating a fractional modeling approach were

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examined in [16], where the authors utilized the Caputo derivative in formulating the model. Fractional-order modeling introduces memory effects into disease dynamics, allowing for a more accurate representation of physical processes. A similar study examining the within-host dynamics of CHIKV using the Caputo modeling approach was conducted in [17]. Additionally, the transmission dynamics of CHIKV between vector and human populations were investigated in [18]. Despite extensive research on CHIKV at the population level, studies focusing on its within-host dynamics remain limited. For instance, [19] proposed a deterministic computational model to analyze CHIKV transmission while incorporating adaptive immune response and latency.

In existing studies on within-host viral dynamics, including those addressing CHIKV, the models are formulated on deterministic assumptions. However, deterministic epidemic models often neglect the inherent randomness present in biological processes such as viral replication, variability in immune responses, and latent cell activation. In realistic biological systems, disease progression and immune interactions rarely follow strictly deterministic patterns; instead, they are shaped by intrinsic randomness and unpredictable fluctuations. As a result, relying solely on deterministic dynamics may fail to capture the true complexity of within-host infection processes. Neglecting these stochastic effects can significantly reduce the reliability of epidemic models, thereby limiting their ability to accurately represent real-world infection dynamics [20]. In contrast, stochastic models address these limitations by incorporating random fluctuations, which become particularly important when the population of infected cells is small, such as during the early stages of infection or during the reactivation of latent reservoirs [21].

Various numerical schemes, including Euler-Maruyama, Runge-Kutta, and Milstein-type methods, have been used in the literature to analyze epidemic models with stochasticity [22,23]. These methods generally offer low-order accuracy and require very small time steps to ensure stability and convergence. In contrast, the Legendre Spectral Collocation method is highly efficient for nonlinear problems due to its use of orthogonal polynomial basis functions and collocation at optimal Legendre-Gauss-Lobatto points [24,25]. Despite the extensive use of conventional numerical schemes, recent advancements highlight the growing interest in Artificial Neural Networks (ANNs) for approximating and analyzing complex dynamical systems arising in science and engineering, including epidemic models [26–29]. A recent study on exponential synchronization for coupled NNs and hybrid delays along with stochastically distributed delayed impulses was presented in [30]. A mathematical modeling approach for the global asymptotic synchronization of non-integer order multi-delay coupled NNs using hybrid control schemes was studied in [31]. A nonlinear non-integer order neural network with uncertainties, perturbations, and delays was studied in [32]. Stochastic neural network architectures have demonstrated superior capability in capturing the uncertainty and variability inherent in complex real-world systems [33]. Their flexibility makes them particularly effective for modeling and control tasks in fields such as engineering and robotics, where system dynamics are often influenced by noise, random perturbations, and nonlinear interactions [34,35]. Feedforward Neural Networks (FFNNs) constitute a fundamental ANN where connections between nodes follow a unidirectional flow, preventing cyclic links. The implementation of FFNN-based structure can be extensively found in machine learning and data analysis. A FFNN is structurally organized into three main components, which include an input layer that receives data, one or more hidden layers that process and extract underlying features, and an output layer responsible for producing the final prediction or classification. Neural network based models have been extensively applied in computational biology owing to their adaptability and strong capacity to approximate nonlinear relationships that are intrinsic to biological systems [36]. Meanwhile, various conventional numerical methods, including spectral techniques, have been developed and employed for simulating nonlinear differential equations [37,38]. In addition to these traditional approaches, NNs serve as efficient computational solvers for differential equation models [39]. Specifically, an ANN is integrated into the differential equation, where it is trained using learned solution values. Once trained, the model can autonomously predict solution values with high accuracy [40].

This study introduces a novel stochastic extension of the within-host CHIKV model with latency cells [19]. Additionally, we develop an efficient computational technique for simulating the stochastic epidemic model using LSCM to analyze CHIKV dynamics under stochastic effects. Furthermore, we present an integrating computing approach providing deeper insights into the complex dynamics of within-host CHIKV transmission.

### 1.1. Novelty and key contributions of the study

Although numerous deterministic mathematical models have been proposed to analyze the within-host transmission dynamics of CHIKV, such approaches often neglect random fluctuations and latent cellular reservoirs, which are crucial for realistic disease progression. In this study, we develop a more robust and biologically relevant framework by incorporating stochastic effects and latent cells into the within-host dynamics of CHIKV. To the best of our knowledge, this is the first study to investigate the combined impact of stochasticity, random fluctuations, and latency on the dynamical behavior of CHIKV using a nonlinear stochastic epidemic modeling approach. Furthermore, unlike conventional numerical methods, we present the first attempt at designing a novel stochastic neuro-computational paradigm for solving within-host stochastic CHIKV models that include latency and adaptive immune responses. Beyond CHIKV, we believe the techniques introduced here can be extended to a wide range of complex nonlinear stochastic epidemic models.

The key contributions of this research are summarized as follows:

- **Intelligent Surrogate Modeling:** For the first time, a stochastic epidemic model is formulated to analyze the within-host dynamics of CHIKV, explicitly incorporating latency and adaptive immune response under varying degrees of randomness.
- **Fast and Accurate Numerical Baseline:** The stochastic CHIKV model is time-integrated using an efficient Legendre Spectral Collocation Method (LSCM), which provides highly accurate approximations and serves as a benchmark for further analysis.
- **Forecasting Model Dynamics:** The evolution of various cell populations is examined through variation in sensitive parameters values and Wiener process intensity, demonstrating how stochastic fluctuations influence CHIKV dynamics.
- **Expertly Tuned FFNN:** Reference solutions generated from LSCM are used to train FFNN. A hybrid Levenberg Marquardt optimization algorithm with hyperparameter tuning is employed to ensure stability and strong generalization capability.
- **Accurate Predictions and Forecasts:** The proposed FFNN framework achieves consistently low prediction errors in the range of $10^{-2}$ to $10^{-6}$ across both single-step and multi-step forecasting horizons.

The subsequent structure of this study is as follows:

Section 2 presents the construction of the within-host CHIKV model, explicitly incorporating stochastic effects. Section 3 provides the fundamental analytical results of the proposed model. Section 4 introduces the numerical scheme based on the LSCM and illustrates its performance through detailed simulations. Section 5 develops the NN-based computational framework and compares its performance with the reference LSCM results. Finally, Section 6 concludes the study.

## 2. Model formulation with latent cell class

The CHIKV compartmental model is first introduced in its deterministic form, incorporating a latent cell class and an adaptive immune

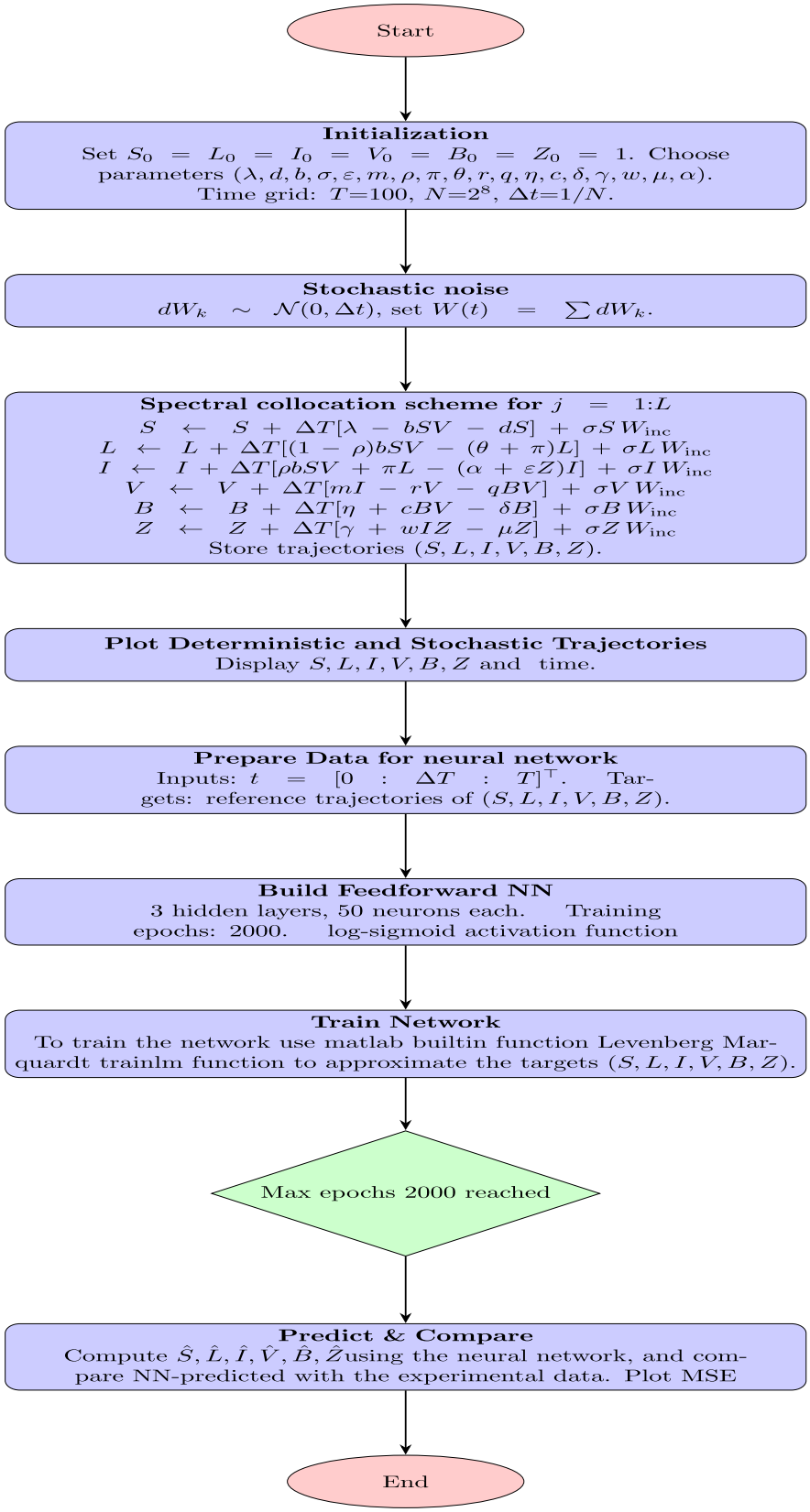


**Fig. 1.** Flowchart of the hybrid LSCM-FFNN algorithm for the stochastic CHIKV model.

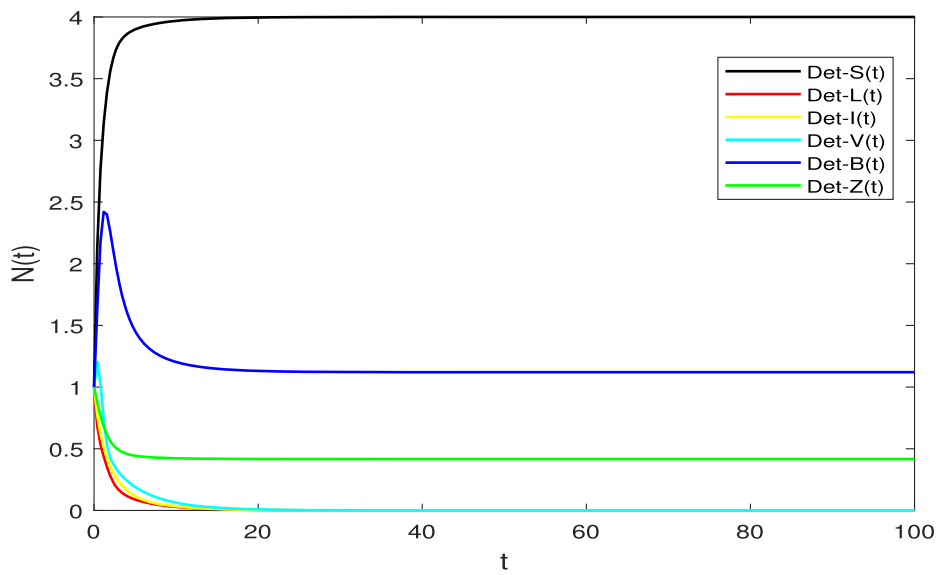


**Fig. 2.** Simulation resulting the dynamics of only deterministic model (1), in the case $R_0 < 1$.

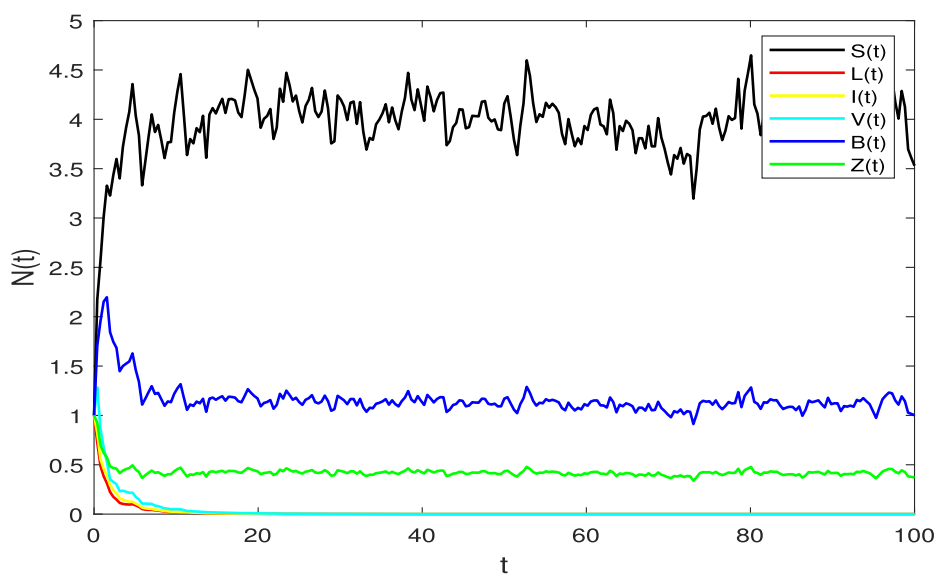


**Fig. 3.** Simulation resulting the dynamics of only the Stochastic model (2), for the case $\bar{\mathcal{R}}_0 < 1$.

response to assess infection dynamics and control. The deterministic model is then extended to a stochastic framework to account for the inherent randomness in CHIKV dynamics under fluctuating environment. Compared to the deterministic approach, the stochastic technique is considered more effective, especially for modeling short-term disease data. Stochastic modeling approach provides a more precise depiction of the underlying dynamics. The deterministic framework is characterized by the subsequent system

$$\begin{cases} \frac{dS(t)}{dt} = \lambda - bS(t)V(t) - dS(t), \\ \frac{dL(t)}{dt} = (1-\rho)bS(t)V(t) - (\theta+\pi)L(t), \\ \frac{dI(t)}{dt} = \rho bS(t)V(t) + \pi L(t) - a(I(t) + \epsilon Z(t))I(t), \\ \frac{dV(t)}{dt} = mI(t) - qV(t)B(t) - rV(t), \\ \frac{dB(t)}{dt} = \eta - \delta B(t) + cB(t)V(t), \\ \frac{dZ(t)}{dt} = \gamma - \mu Z(t) + wI(t)Z(t), \end{cases} \tag{1}$$

with the initial conditions listed below:

$$S(0) = S_0 > 0, L(0) = L_0 \geq 0, I(0) = I_0 \geq 0, V(0) = V_0 \geq 0,$$
$$B(0) = B_0 \geq 0, Z(0) = Z_0 \geq 0.$$

In the formulation of model (1), the state variables $S(t)$, $L(t)$, $I(t)$, $V(t)$, $B(t)$, and $Z(t)$ denote the concentrations of susceptible (uninfected) cells, latent cells, infected cells, CHIKV particles, antibodies, and cytotoxic T lymphocyte (CTL) cells within the human body, respectively. In the model (1) we incorporate multiple parameters, where $d$ denotes the natural death rate constant of uninfected cells, and $\lambda$ represents their constant production rate. Susceptible cells become infected at an incidence rate of $b$, with a fraction $\rho$ transitioning directly into the actively infected cell compartment at a rate of $b\rho SV$, while the remaining fraction enters the latent class $L$. The parameter $a$ denotes the rate at which the actively infected cells die out, $\epsilon$ represents the CHIKV elimination rate due to CTL immune responses, and $\pi$ signifies the transition rate from latent to actively infected cells. The average count and decay rate of

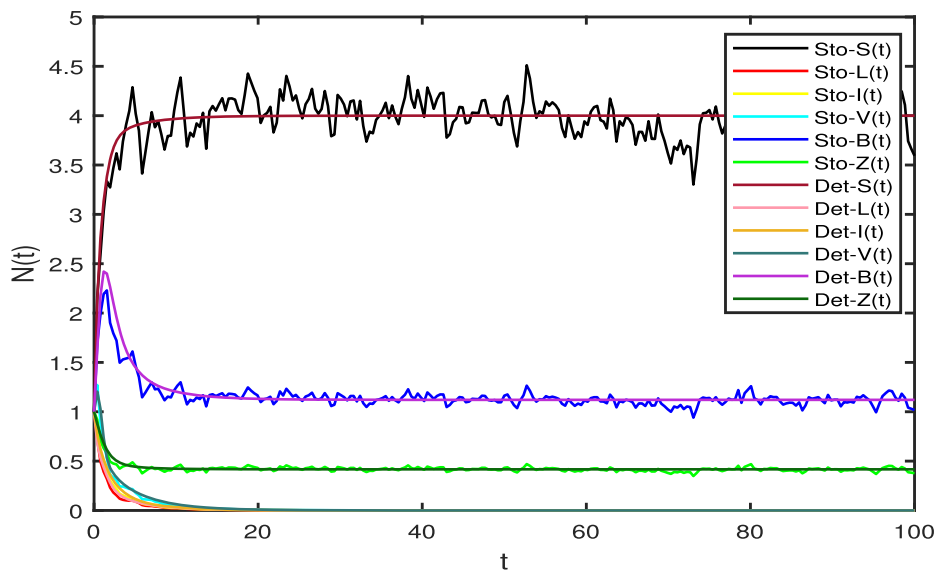


**Fig. 4.** Comparisonof deterministic case model (1) and stochastic case model (2), in the case when both threshold parameters $\mathcal{R}_0 < 1$ and $\bar{\mathcal{R}}_0 < 1$.

**Table 1**
Biological interpretation of the parameters embedded in the CHIKV model.

| Parameter | Biological description |
|---|---|
| $\lambda$ | Recruitment rate of uninfected cells |
| $d$ | Death rate of uninfected cell |
| $b$ | Contact rate |
| $\rho$ | Fraction of uninfected cells to join actively infected cells |
| $\theta$ | Death rate of latent cells |
| $\pi$ | Transition from latent to infected cells class |
| $a$ | Decay rate of infected cells |
| $\epsilon$ | CHIKV elimination rate |
| $m$ | Average count of CHIKV particles |
| $q$ | Antibodies attack rate at the virus |
| $r$ | Decay rate of CHIKV particles |
| $\eta$ | The number of antibodies in blood concentration |
| $\delta$ | Decay rate of antibodies |
| $c$ | The increasing rate of antibodies |
| $\mu$ | Decay rate of CTL cells |
| $\gamma$ | Average count of CTL cells |
| $\omega$ | The enhancement rate of CTL cells |

CHIKV particles are represented by $m$ and $r$ respectively. The parameter $q$ denotes the rate at which antibodies attack the virus, while $\eta$ represents the number of antibodies in blood circulation. The rate of antibody production is given by $c$, and their decay rate is $\delta$. Similarly, the average count of CTL cells is denoted by $\gamma$, with $\omega$ representing their proliferation rate and $\mu$ indicating their death rate. In addition to this, the biological description of the of parameters are mentioned in Table 1.

### 2.1. Extension with stochastic effects

Although substantial progress has been made in computational modeling of CHIKV dynamics, most existing studies still rely primarily on deterministic frameworks. These models assume fixed parameters and generate predicted outcomes accordingly. However, actual biological systems exhibit inherent randomness and variability, which can be fully captured using deterministic models. Therefore, to address this limitation, we present an innovative extension of the classical system (1) describing the CHIKV within host dynamics by integrating the stochastic effects. This enhancement offers a more realistic depiction of CHIKV dynamics, better reflecting the unpredictable nature of biological processes.

For this purpose we reconstruct the deterministic CHIKV within-host model (1) to the following nonlinear stochastic model:

$$\begin{cases} \frac{dS(t)}{dt} = \lambda - bS(t)V(t) - dS(t) + \sigma_1 S(t)dW_1(t), \\ \frac{dL(t)}{dt} = (1-\rho)bS(t)V(t) - (\theta+\pi)L(t) + \sigma_2 L(t)dW_2(t), \\ \frac{dI(t)}{dt} = \rho bS(t)V(t) + \pi L(t) - (a+\epsilon Z(t))I(t) + \sigma_3 I(t)dW_3(t), \\ \frac{dV(t)}{dt} = mI(t) - qV(t)B(t) - rV(t) + \sigma_4 V(t)dW_4(t), \\ \frac{dB(t)}{dt} = \eta - \delta B(t) + cV(t)B(t) + \sigma_5 B(t)dW_5(t), \\ \frac{dZ(t)}{dt} = \gamma - \mu Z(t) + wI(t)Z(t) + \sigma_6 Z(t)dW_6(t). \end{cases} \tag{2}$$

In the above system (2), $dW_i(t)$ $i = 1, \dots, 6$, denote a random process characterized by erratic dynamics and known as the Wiener process or Brownian motion. $dW(t)$ is normally distributed with mean 0 and variance $dt$, $dW \sim \mathcal{N}(0, dt)$. The parameter $\sigma_i$ such as $\sigma_1$, $\sigma_2$, $\sigma_3$, $\sigma_4$, $\sigma_5$, $\sigma_6$ quantifies the degree of randomness or volatility in each differential equation of the system. Larger values of $\sigma$ the greater influence of stochastic effects, such as environmental fluctuations, random interactions, or other unpredictable factors.

## 3. Basic analysis of the stochastic model

Prior to the numerical analysis, we provide an overview of the fundamental mathematical properties of the within-host CHIKV model.

### 3.1. Threshold number ($R_0$) of model (1)

We evaluate the reproduction number of the SLIVBZ model (2), which accurately characterizes the within-cells CHIKV dynamics. To determine the threshold number of the model (1) for the deterministic case, we analyze the following subsystem comprising the key infection dynamics:

$$\begin{cases} \frac{dL(t)}{dt} = (1-\rho)bS(t)V(t) - (\theta+\pi)L(t), \\ \frac{dI(t)}{dt} = bS(t)V(t) - \epsilon I(t)Z(t) - aI(t), \\ \frac{dV(t)}{dt} = mI(t) - qV(t)B(t) - rV(t), \\ \frac{dZ(t)}{dt} = \gamma - \mu Z(t) + wI(t)Z(t). \end{cases} \tag{3}$$

By using the well-known next generation technique, the expression for $R_0$ of the deterministic model is defined by

$$R_0 = \frac{bm\delta\lambda\mu(\pi+\theta\rho)}{d(r\delta+q\eta)(\theta+\pi)(a\mu+\epsilon\gamma)}. \tag{4}$$

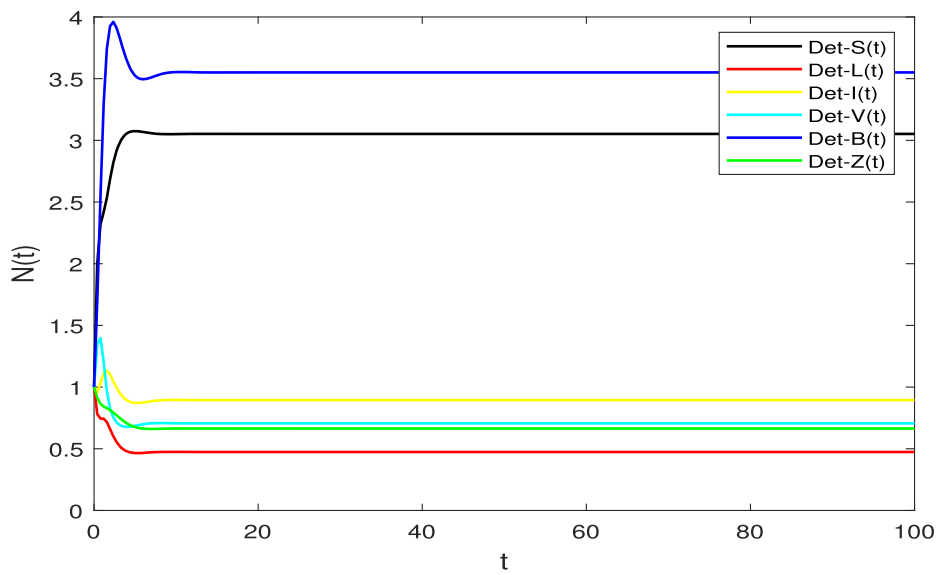


**Fig. 5.** Simulation resulting the dynamics of only deterministic system Eq. (1), where $\mathcal{R}_0 > 1$.

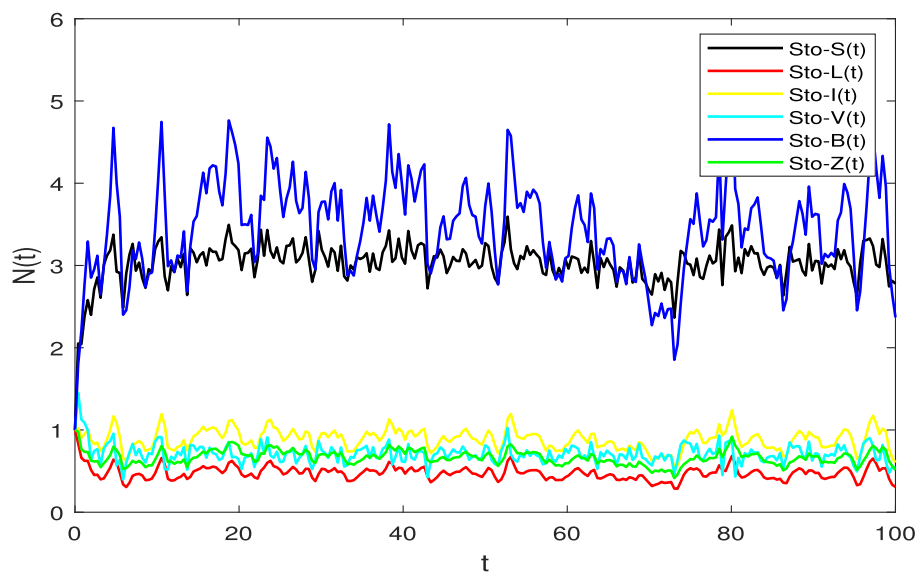


**Fig. 6.** Simulation resulting the dynamics of only stochastic (2), where $\bar{\mathcal{R}}_0 > 1$.

**Theorem 1.** *If $R_0 < 1$ then there exists an CHIKV-free equilibrium $\left(S^*(t), L^*(t), I^*(t), V^*(t), B(t), Z^*(t)\right) \to \left(\frac{\lambda}{d}, 0, 0, 0, \frac{\eta}{\delta}, \frac{\gamma}{\mu}\right)$. Furthermore, if $R_0 > 1$, then system (1) has a unique and stable endemic equilibrium (EE)*

$$E^* = \left\{S^*(t), L^*(t), I^*(t), V^*(t), B^*(t), Z^*(t)\right\},$$

*where,*

$$S^*(t) = \frac{(r+qB)(a+\epsilon Z)}{bm}, \quad L^*(t) = \frac{(1-\rho)bSV}{\theta+\pi}, \quad I^*(t) = \frac{(r+qB)V}{m},$$
$$B^*(t) = \frac{\eta}{\delta - cV}, \quad Z^*(t) = \frac{m\gamma}{m\mu - \omega V(r+qB)}.$$

**Proof.** The proof can be obtained by solving system (1) at CHIKV-free and endemic states. Moreover, the detail proof is provided in [19]. □

**Lemma:** The solution of Eq. (2), $(S(t), L(t), I(t), V(t), B(t), Z(t))$ with each initial condition $(S(0), L(0), I(0), V(0), B(0), Z(0)) \in R_6^+$: has the subsequent properties,

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t \frac{dS(s)}{N(t)} dW(s) = 0, \tag{5}$$

The threshold number $\bar{R}_0$ for the model (2) in stochastic case is formulated as follows:

$$\bar{R}_0 = R_0 - \left\{ \frac{\rho a\lambda\mu + \rho\lambda\epsilon\gamma}{\mu b} - \frac{\sigma^2\lambda^2 a\mu + \epsilon\gamma}{a^2\mu b^2} \right\}.$$

**Theorem 2.** *For the stochastic CHIKV compartmental model (2), if $\sigma^2 > \max\left\{\frac{\rho ba}{\lambda}, \frac{\mu b^2}{a\mu+\epsilon\gamma}\right\}$ or $\sigma^2 < \frac{\rho ba}{\lambda}$ with $\bar{\mathcal{R}}_0 < 1$, then this implies the extension of the infected class.*

**Proof.** Let $(S(t),\ L(t),\ I(t),\ V(t),\ B(t),\ Z(t))$ be the solution of model (2) with nonnegative initial values in $R_+^6$, the with the help of Itô formula we proceed as

$$d\ln I(t) = \left(\rho abS(t) - (a+\epsilon Z(t)) - \frac{\sigma^2 S^2(t)}{2}\right)dt + \sigma_3 I(t)dW_3(t). \tag{6}$$

Integration over the [0, t], result as follows:

$$\ln I(t) = \ln I(0) + \int_0^t \left(a\rho bS(t) - (a+\epsilon Z(t)) - \frac{\sigma^2 S^2(t)}{2}\right)dt + \int_0^t \sigma_3 I(t)dW_3(t). \tag{7}$$

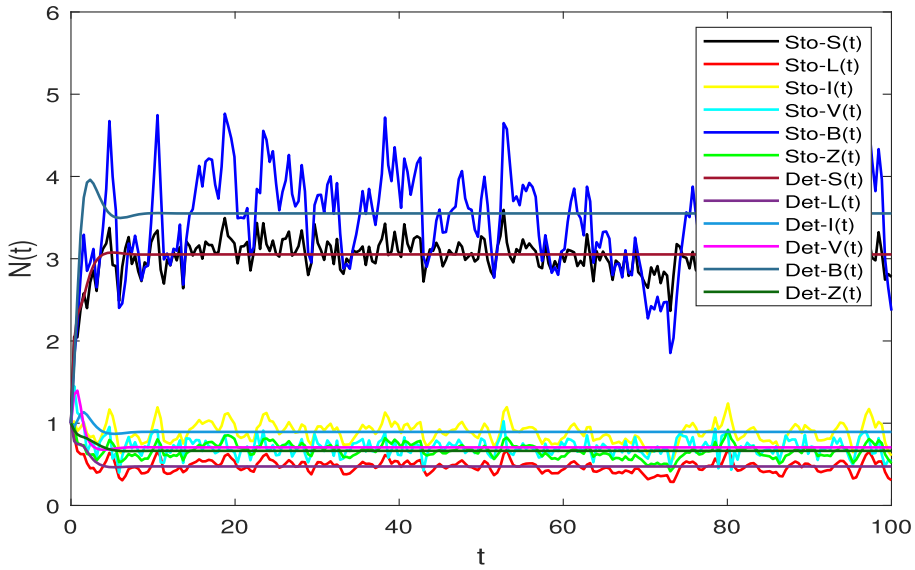


**Fig. 7.** Simulation demonstrating comparison of the models (1) and (2), when $\mathcal{R}_0 > 1$ and $\bar{\mathcal{R}}_0 > 1$.

Here we discuss two cases, if $\sigma^2 > \frac{\rho b a}{\lambda}$, then we get from Eq. (7):

$$\ln I(t) \le \ln I(0) + \left(\frac{\rho b^2}{\sigma^2} - \frac{\mu b^2}{a\mu + \epsilon\gamma}\right)t + \int_0^t \sigma_3 I(t) dW_3(t). \tag{8}$$

Dividing (8) by $t > 0$, we have

$$\frac{\ln I(t)}{t} \le \frac{1}{t}\ln I(0) + \left(\frac{\rho b^2}{\sigma^2} - \frac{\mu b^2}{a\mu + \epsilon\gamma}\right)t + \frac{1}{t}\int_0^t \sigma_3 I(t) dW_3(t). \tag{9}$$

Taking $\lim_{t\to\infty}$ along with Lemma 3, Eq. (9) gives:

$$\lim_{t\to\infty} \frac{\ln I(t)}{t} \le -\left(\frac{\mu b^2}{a\mu + \epsilon\gamma} - \frac{\rho b^2}{\sigma^2}\right) < 0.$$

When $\sigma^2 > \frac{\mu b^2}{a\mu+\epsilon\gamma}$, implies that $\lim_{t\to\infty} I(t) = 0$. If $\sigma^2 < \frac{\rho b a}{\lambda}$, as Eq. (7) become:

$$\ln I(t) \le \ln I(0) + \left(\frac{\rho b\lambda}{a} - \frac{\sigma^2\lambda^2}{a^2} - \frac{\mu b^2}{a\mu + \epsilon\gamma}\right)t + \int_0^t \sigma_3 I(t) dW_3(t). \tag{10}$$

Dividing Eq. (10) by $t > 0$ we get:

$$\frac{1}{t}\ln I(t) \le \frac{1}{t}\ln I(0) + \left(\frac{\rho b\lambda}{a} - \frac{\sigma^2\lambda^2}{a^2} - \frac{\mu b^2}{a\mu + \epsilon\gamma}\right)t + \frac{1}{t}\int_0^t \sigma_3 I(t) dW_3(t). \tag{11}$$

Using the aforementioned Lemma and apply limit over (16) which leads to

$$\lim_{t\to\infty} \frac{\ln I(t)}{t} = \frac{\mu b^2}{a\mu + \gamma\epsilon}\left(\frac{\rho b\lambda(a\mu + \epsilon\gamma)}{\mu b^2} - \frac{\sigma^2\lambda^2(a\mu + \epsilon\gamma)}{a^2\mu b^2} - 1\right). \tag{12}$$

$$\lim_{t\to\infty} \frac{\ln I(t)}{t} = \frac{\mu b^2}{a\mu + \gamma\epsilon}\left(\frac{\rho a\lambda\mu + \rho\lambda\epsilon\gamma}{\mu b} - \frac{\sigma^2\lambda^2 a\mu + \epsilon\gamma}{a^2\mu b^2} - 1\right). \tag{13}$$

$$\lim_{t\to\infty} \frac{\ln I(t)}{t} \le \frac{\mu b^2}{a\mu + \epsilon\gamma}(\bar{\mathcal{R}}_0 - 1).$$

Thus for $\bar{\mathcal{R}}_0$ less than one it implies

$$\lim_{t\to\infty} \frac{\ln I(t)}{t} < 0.$$

Hence,

$$\lim_{t\to\infty} I(t) = 0.$$

□

## 4. Numerical scheme

### 4.1. Spectral method

We first present an overview of the Legendre polynomials approximation implemented by numerous authors in the recent literature [37,38]. For a function $g(t)$ with $t \in [-1, 1]$; then utilizing the spectral scheme, we have the following approximation

$$g(t) = \sum_{\iota=0}^{m} g_\iota \mathrm{K}_\iota(t), \tag{14}$$

where, $\mathrm{K}_n(t)$ is $nth$ order Legendre polynomial, $g_i$ are unknowns coefficients, such that the nodes fulfil, $-1 = t_0 < \ldots < t_m = 1$. Moreover,

$$\mathrm{K}_\iota(t) = \frac{1}{2^\iota}\sum_{j=0}^{\left[\frac{\iota}{2}\right]} (-1)^j \binom{\iota}{j}\binom{2\iota - 2j}{\iota} t^{\iota-2j}; \quad (\iota = 0, \ldots, m) \tag{15}$$

$$\left[\frac{j}{2}\right] = \begin{cases} \frac{\iota}{2}, & if \quad \iota \quad \text{even}, \\ \frac{\iota-1}{2}, & if \quad \iota \quad \text{odd}. \end{cases}$$

Further, the Legendre-Gauss-Lobatto nodes are taken in consideration $\{t_\iota\}_{\iota=0}^{N}$. Integration of Eq. (2), on the closed interval $[0, t]$ lead to

$$\begin{cases} S(t) = & S_0 + \int_0^t \left(\lambda - dS(r) - bS(r)V(r)\right)dr + \int_0^t \sigma_1 S(r) dW_1(r), \\ L(t) = & L_0 + \int_0^t \left((1-\rho)bS(r)V(r)\right)dr - (\theta + \pi)L(r)dr + \int_0^t \sigma_2 I(r) dW_2(r), \\ I(t) = & I_0 + \int_0^t \left(\rho bS(r)V(r) + \pi L(r) - aI(r) - \epsilon I(r)Z(r)\right)dr \\ & + \int_0^t \sigma_3 I(r) dW_3(r), \\ V(t) = & V_0 + \int_0^t \left(mI(r) - rV(r) - qV(r)B(r)\right)dr + \int_0^t \sigma_4 V(r) dW_4(r), \\ B(t) = & B_0 + \int_0^t \left(\eta + cB(r)V(r) - \delta B(r)\right)dr + \int_0^t \sigma_5 B(r) dW_5(r), \\ Z(t) = & Z_0 + \int_0^t \left(\gamma + wI(r)Z(r) - \mu Z(r)\right)dr + \int_0^t \sigma_6 Z(r) dW_6(r). \end{cases} \tag{16}$$

Using a linear transformation $r = \frac{t}{2}(v_0 + 1)$, we investigate the LSCM over the closed interval [-1,1], as a result Eq. (16) becomes:

$$S(t) = S_0 + \int_{-1}^{1} \left(\lambda - dS\left(\frac{t}{2}(v_0+1)\right) - bS\left(\frac{t}{2}(v_0+1)\right)V\left(\frac{t}{2}(v_0+1)\right)\right)$$
$$\times d\left(\frac{t}{2}(v_0+1)\right) + \int_{-1}^{1} \sigma_1 S\left(\frac{t}{2}(v_0+1)\right) dW_1\left(\frac{t}{2}(v_0+1)\right),$$
$$L(t) = L_0 + \int_{-1}^{1} \left((1-\rho)bS\left(\frac{t}{2}(v_0+1)\right)V\left(\frac{t}{2}(v_0+1)\right) - (\theta+\pi)L\left(\frac{t}{2}(v_0+1)\right)\right)$$

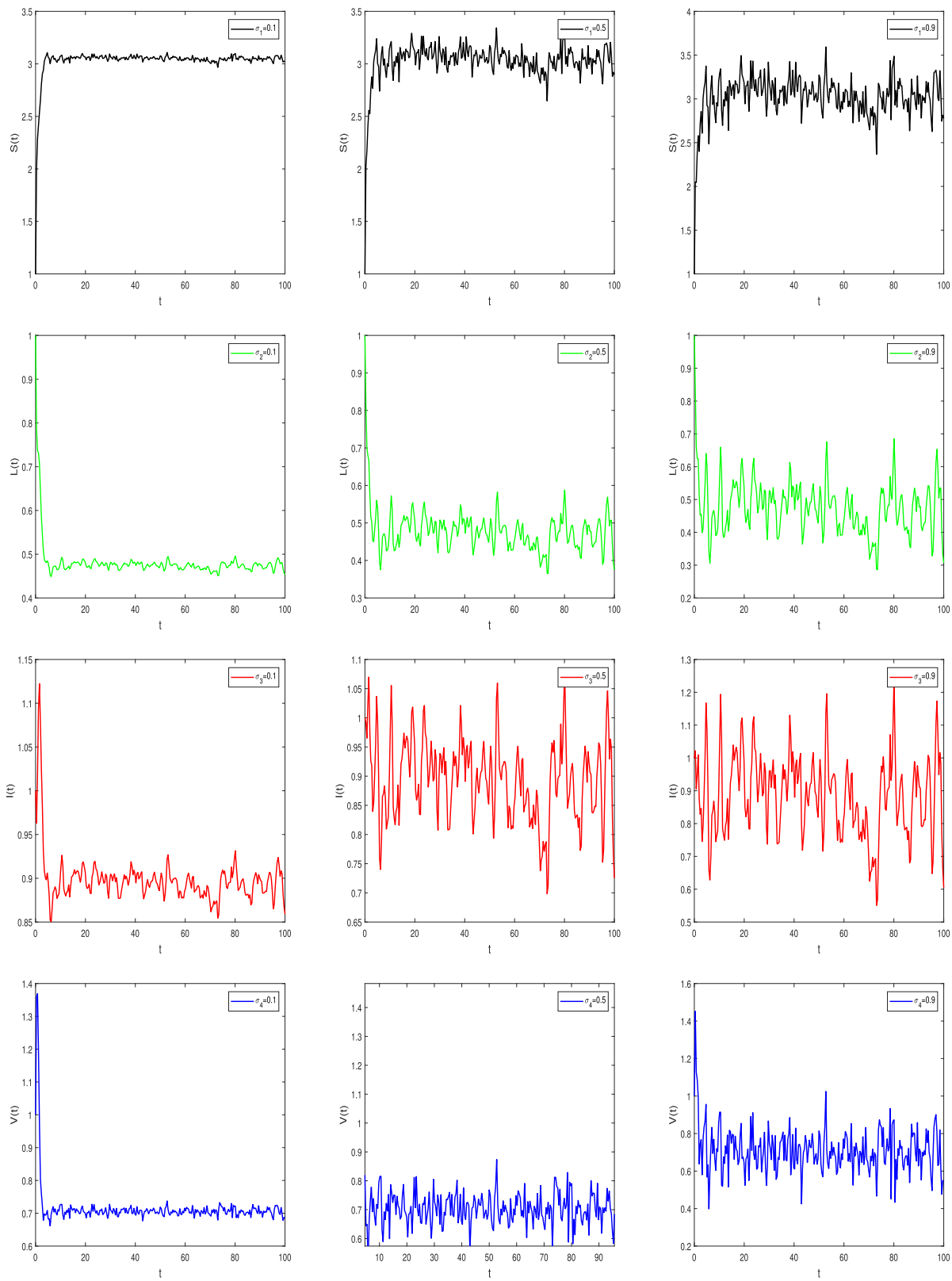


**Fig. 8.** Simulation illustrating the impact of $\sigma_1, \sigma_2, \sigma_3, \sigma_4$, with values 0.1, 0.5 0.9.

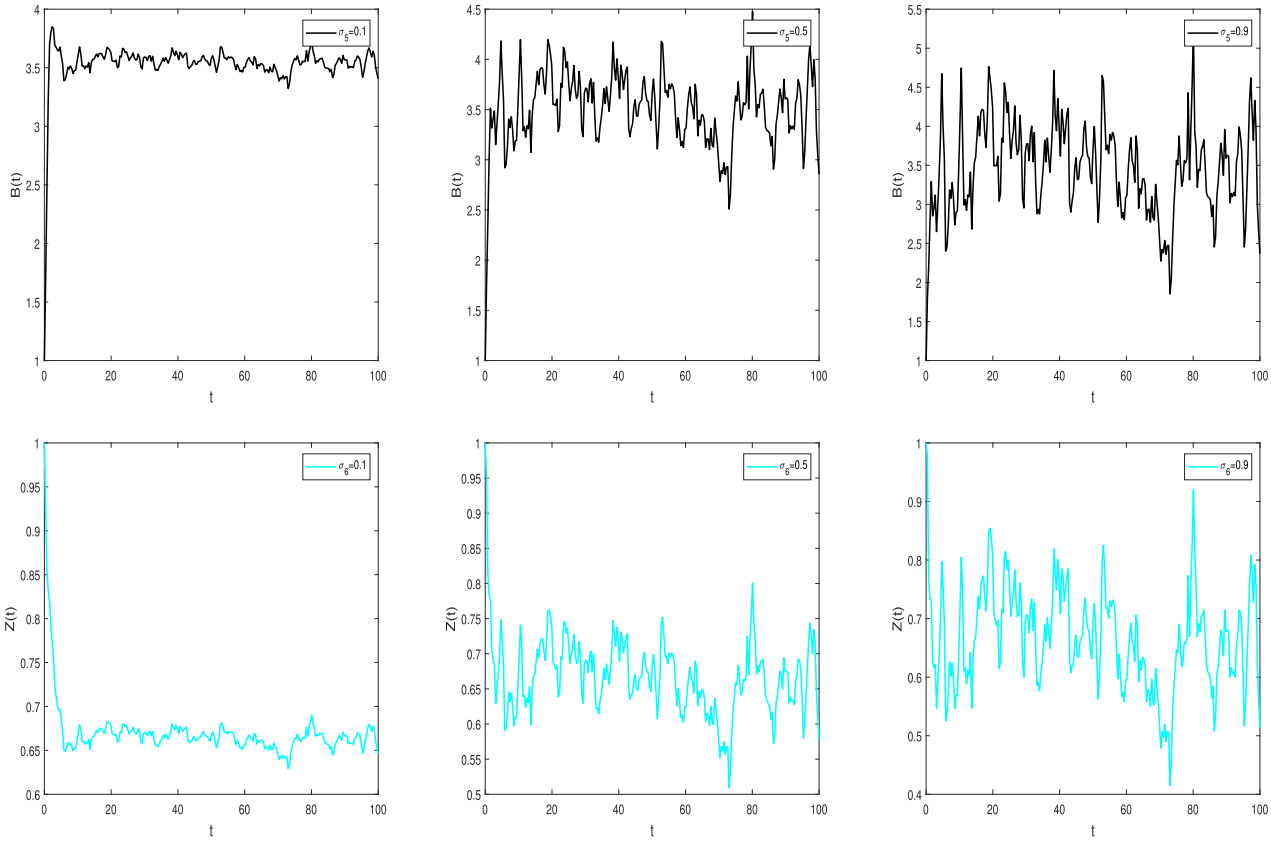


**Fig. 9.** Simulation illustrating the impact of $\sigma_5, \sigma_6$, with rates 0.1, 0.5 0.9.

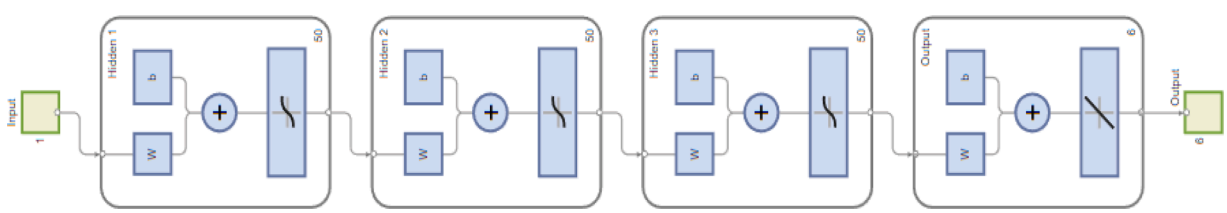


**Fig. 10.** The NN design implemented in this investigation.

$$\times d\Big(\frac{t}{2}(v_0+1)\Big)+\int_{-1}^{1}\sigma_2 I\Big(\frac{t}{2}(v_0+1)\Big)dW_2\Big(\frac{t}{2}(v_0+1)\Big),$$

$$I(t)=I_0+\int_{-1}^{1}\Big(\rho bS\Big(\frac{t}{2}(v_0+1)\Big)V\Big(\frac{t}{2}(v_0+1)\Big)+\pi L\Big(\frac{t}{2}(v_0+1)\Big)$$
$$-aI\Big(\frac{t}{2}(v_0+1)\Big)-\epsilon I\Big(\frac{t}{2}(v_0+1)\Big)Z\Big(\frac{t}{2}(v_0+1)\Big)\Big)d\Big(\frac{t}{2}(v_0+1)\Big)$$
$$+\int_{-1}^{1}\sigma_3 I\Big(\frac{t}{2}(v_0+1)\Big)dW_3\Big(\frac{t}{2}(v_0+1)\Big),$$

$$V(t)=V_0+\int_{-1}^{1}\Big(mI\Big(\frac{t}{2}(v_0+1)\Big)-rV\Big(\frac{t}{2}(v_0+1)\Big)-qV\Big(\frac{t}{2}(v_0+1)\Big)$$
$$\times B\Big(\frac{t}{2}(v_0+1)\Big)\Big)d\Big(\frac{t}{2}(v_0+1)\Big)+\int_{-1}^{1}\sigma_4 V\Big(\frac{t}{2}(v_0+1)\Big)$$
$$\times dW_4\Big(\frac{t}{2}(v_0+1)\Big),$$

$$B(t)=B_0+\int_{-1}^{1}\Big(\eta+cB\Big(\frac{t}{2}(v_0+1)\Big)V\Big(\frac{t}{2}(v_0+1)\Big)-\sigma B\Big(\frac{t}{2}(v_0+1)\Big)\Big)$$
$$\times d\Big(\frac{t}{2}(v_0+1)\Big)+\int_{-1}^{1}\sigma_5 B\Big(\frac{t}{2}(v_0+1)\Big)dW_5\Big(\frac{t}{2}(v_0+1)\Big),$$

$$Z(t)=Z_0+\int_{-1}^{1}\Big(\gamma+wI\Big(\frac{t}{2}(v_0+1)\Big)Z(\frac{t}{2}(v_0+1))-\mu Z\Big(\frac{t}{2}(v_0+1)\Big)\Big)$$
$$\times d\Big(\frac{t}{2}(v_0+1)\Big)+\int_{-1}^{1}\sigma_6 Z\Big(\frac{t}{2}(v_0+1)\Big)dW_6\Big(\frac{t}{2}(v_0+1)\Big). \quad (17)$$

The semi-discrete spectral equations for in Eq. 17 is as follows:

$$S(t)=S_0+\frac{t}{2}\sum_{k=0}^{M}\Big\{\lambda-dS\frac{t}{2}(v_0+1)-bS\frac{t}{2}(v_0+1)V\frac{t}{2}(v_0+1)dr\Big\}W_k$$
$$+\frac{t}{2}\sum_{k=0}^{M}\Big\{\sigma_1 S\frac{t}{2}(v_0+1)dW_1\frac{t}{2}(v_0+1)\Big\}W_k^*,$$

$$L(t)=L_0+\frac{t}{2}\sum_{k=0}^{M}\Big\{(1-\rho)bS(\frac{t}{2}(v_0+1))V(\frac{t}{2}(v_0+1))dr-(\theta+\pi)$$
$$\times L(\frac{t}{2}(v_0+1))dr\Big\}W_k$$
$$+\frac{t}{2}\sum_{k=0}^{M}\Big\{\sigma_2 L\frac{t}{2}(v_0+1)dW_2\frac{t}{2}(v_0+1)\Big\}W_k^*,$$

$$I(t)=I_0+\frac{t}{2}\sum_{k=0}^{M}\Big\{\rho bS\frac{t}{2}(v_0+1)V\frac{t}{2}(v_0+1)+\pi L\frac{t}{2}(v_0+1)$$
$$-aI\frac{t}{2}(v_0+1)-\epsilon I\frac{t}{2}(v_0+1)$$

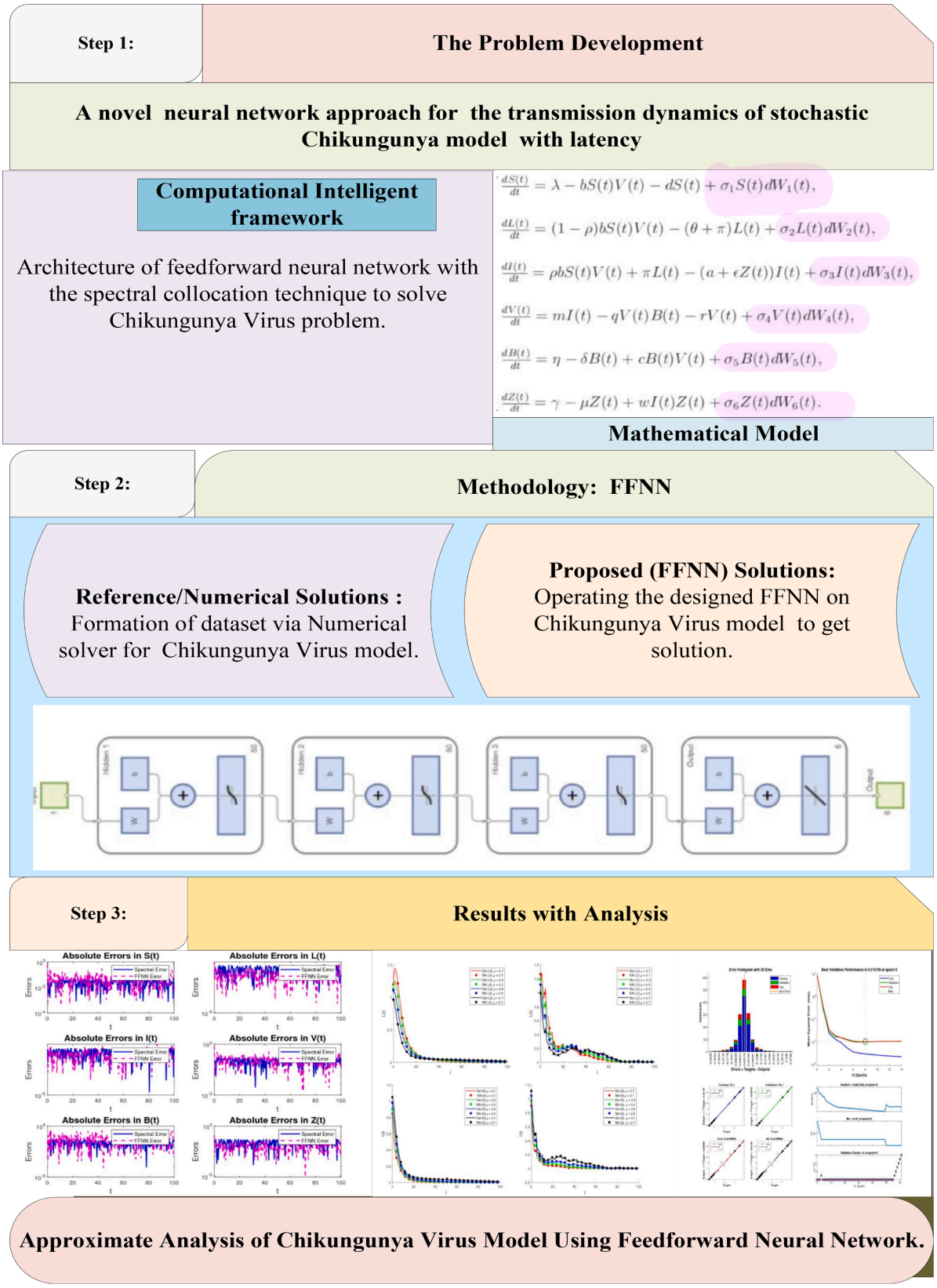


**Fig. 11.** Graphical abstract of the proposed CHIKV disease model.

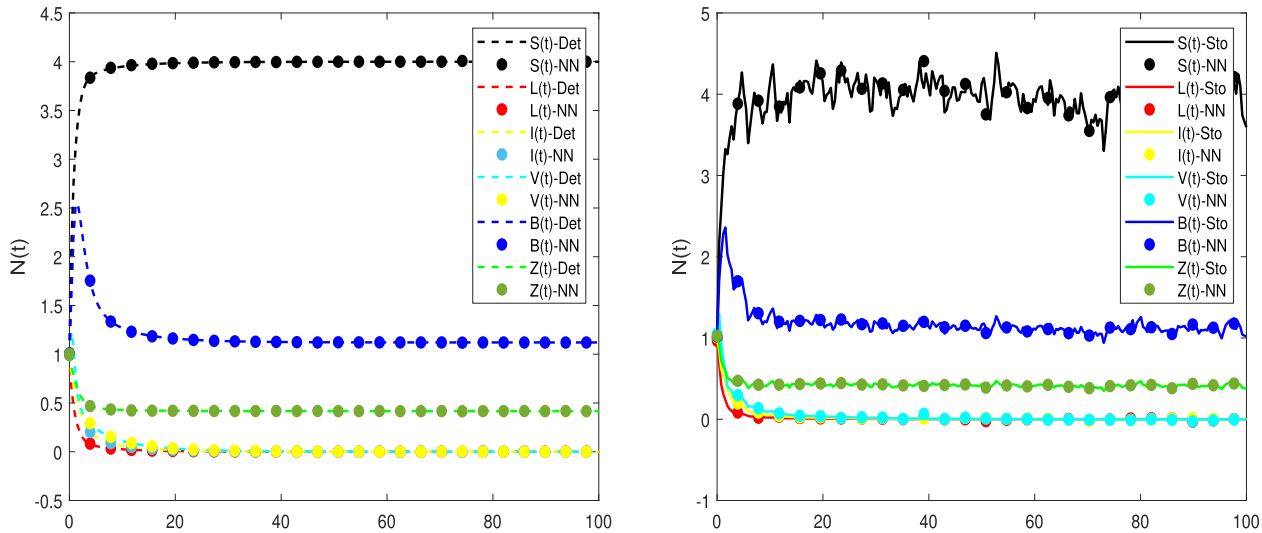


**Fig. 12.** A Comparative analysis of solutions for NN-based, stochastic and deterministic systems for $R_0 < 1$.

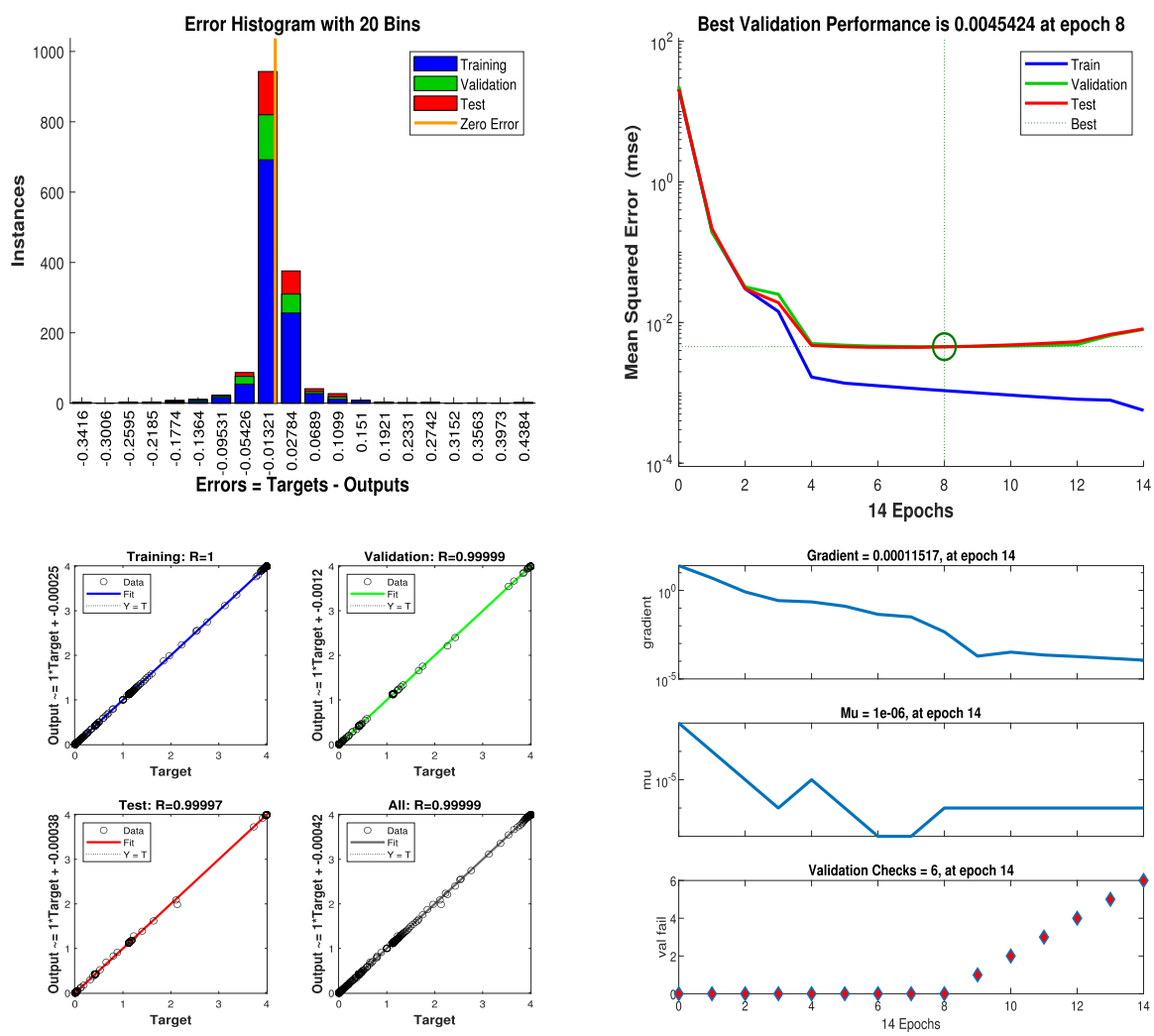


**Fig. 13.** Optimal authentication, error histogram, regression analysis, and gradient outcome for solving the CHIKV model for $R_0 < 1$.

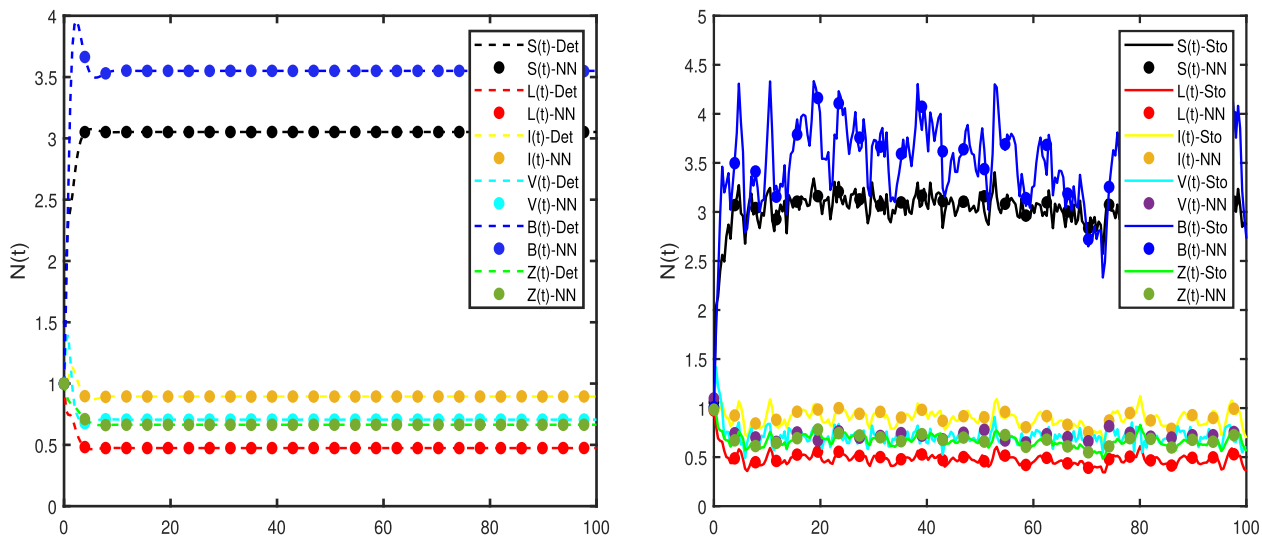


**Fig. 14.** A Comparative analysis of solutions for NN-based, stochastic and deterministic systems for $R_0 > 1$.

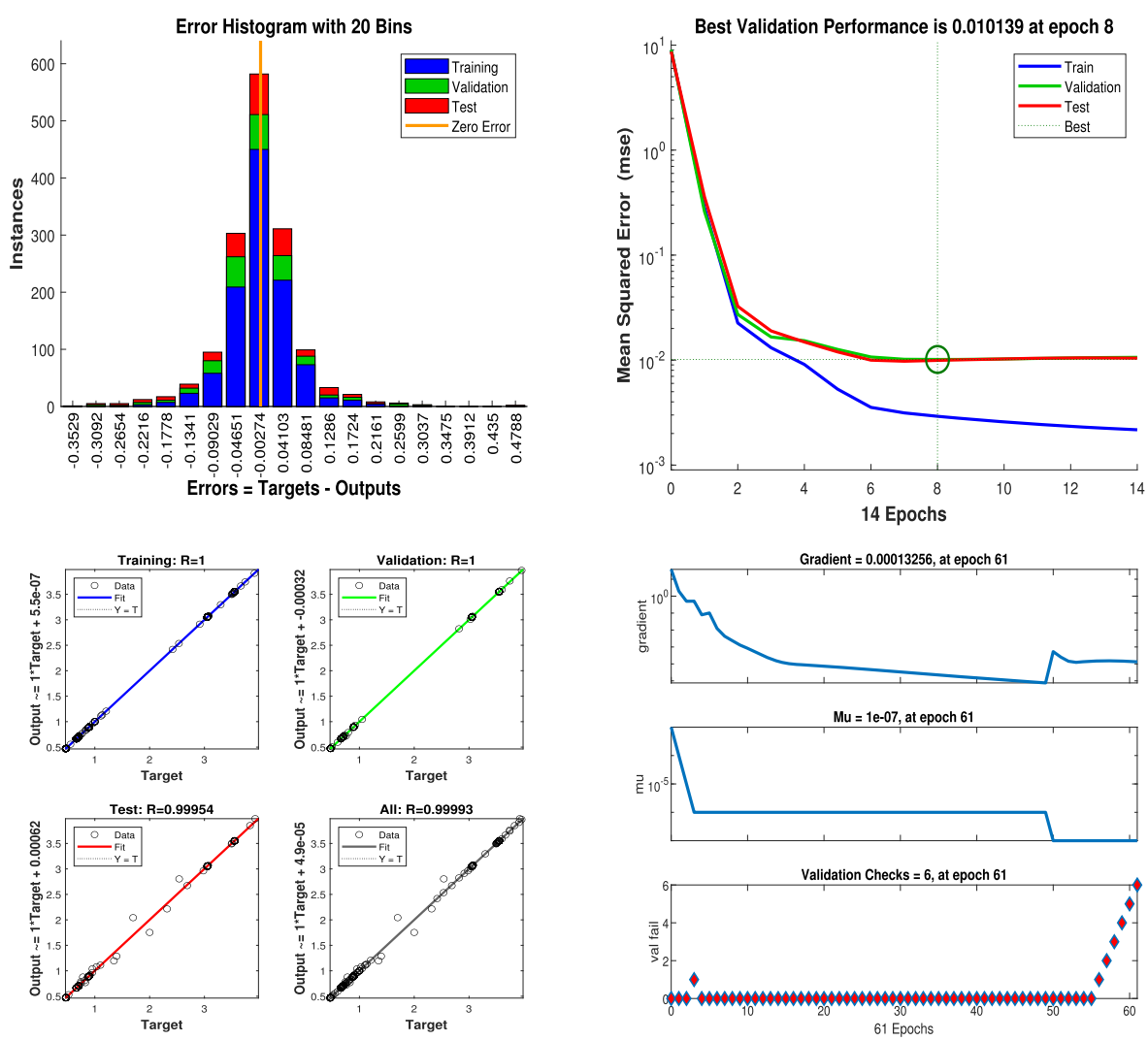


**Fig. 15.** Optimal authentication, error histogram, regression analysis, and gradient outcome for solving the CHIKV model for $R_0 > 1$.

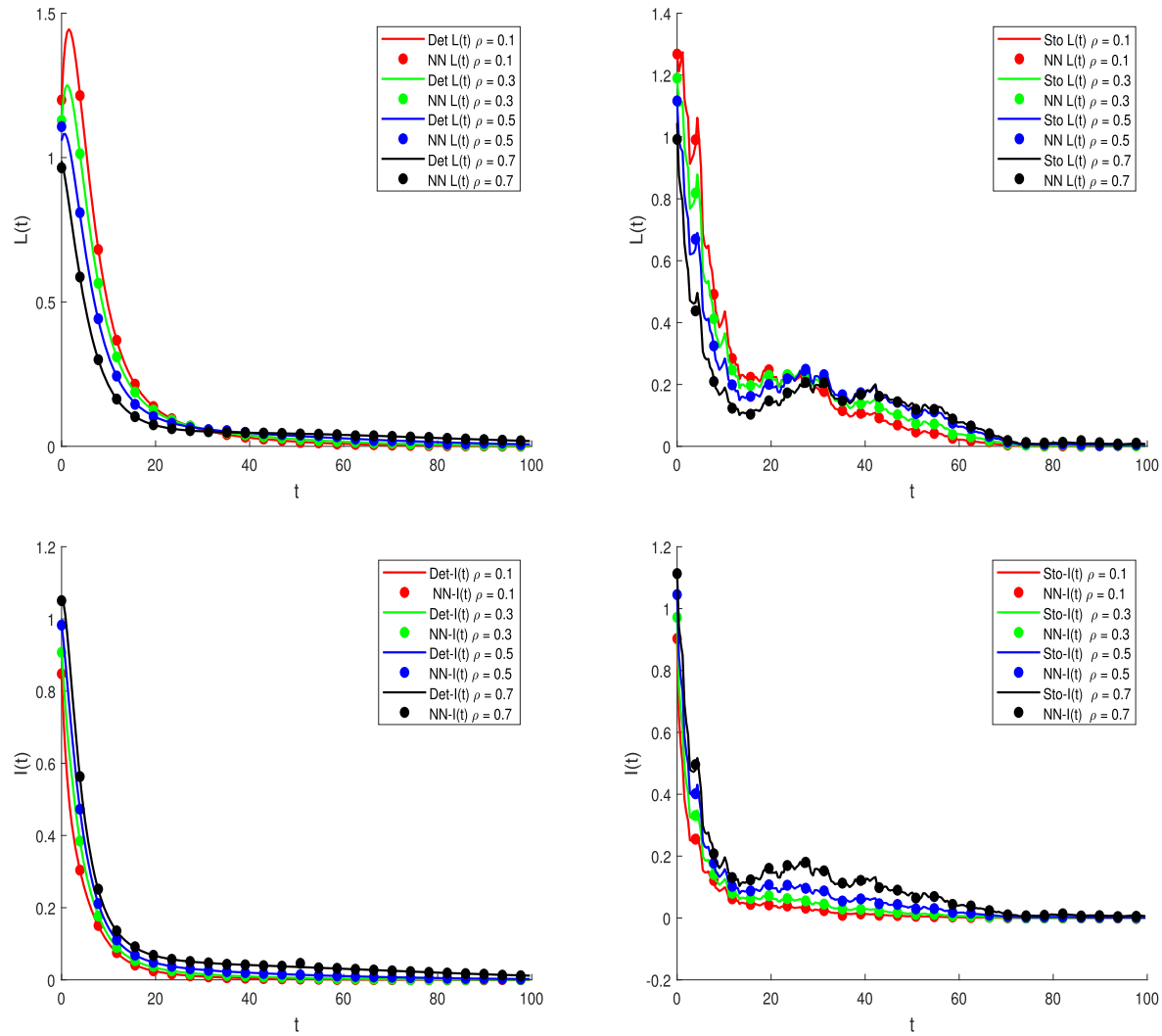


**Fig. 16.** Simulation illustrating the impact of $\rho$ on Latently infected class L(t) and actively infected class $I(t)$.

$$Z\frac{t}{2}(v_0+1)dr\}W_k+\frac{t}{2}\sum_{k=0}^{M}\{\sigma_3 I\frac{t}{2}(v_0+1)dW_3\frac{t}{2}(v_0+1)\}W_k^*,$$

$$\begin{aligned}V(t)=V_0+\frac{t}{2}\sum_{k=0}^{M}\{&mI\frac{t}{2}(v_0+1)-rV\frac{t}{2}(v_0+1)-qV\frac{t}{2}(v_0+1)\\&\times B\frac{t}{2}(v_0+1)dr\}W_k\\&+\frac{t}{2}\sum_{k=0}^{M}\{\sigma_4 V\frac{t}{2}(v_0+1)dW_4\frac{t}{2}(v_0+1)\}W_k^*,\\B(t)=B_0+\frac{t}{2}\sum_{k=0}^{M}\{&\eta+cB\frac{t}{2}(v_0+1)V\frac{t}{2}(v_0+1)-\delta\\&\times B\frac{t}{2}(v_0+1)dr\}W_k\\&+\frac{t}{2}\sum_{k=0}^{M}\{\sigma_5 B\frac{t}{2}(v_0+1)dW_5\frac{t}{2}(v_0+1)\}W_k^*,\\Z(t)=Z_0+\frac{t}{2}\sum_{k=0}^{M}\{&\gamma+wI\frac{t}{2}(v_0+1)Z\frac{t}{2}(v_0+1)-\mu Z\frac{t}{2}(v_0+1)dr\}W_k\\&+\frac{t}{2}\sum_{k=0}^{M}\{\sigma_6 Z\frac{t}{2}(v_0+1)dW_6\frac{t}{2}(v_0+1)\}W_k^*.x\end{aligned}\tag{18}$$

The corresponding weight function is given as

$$W_k=\frac{2}{(1-r_k^2)[L'_{M+1}(r_k)]^2},\qquad 0\le k\le M.$$

A randomized weight function can be utilized in Legendre-Gauss quadrature to obtain an equivalent outcome

$$W_k^*=\sqrt{W_k}\times randam(1,\ M),\qquad 0\le k\le M.$$

For $S(t), L(t), I(t), V(t), B(t), Z(t)$, using Eq. 15 we proceed with the approximations as follows:

$$\begin{aligned}&S(t)=\sum_{\vartheta=0}^{M}S_\vartheta K_\vartheta(t),\qquad L(t)=\sum_{\vartheta=0}^{M}L_\vartheta K_\vartheta(t),\qquad I(t)=\sum_{\vartheta=0}^{M}I_\vartheta K_\vartheta(t),\\&V(t)=\sum_{\vartheta=0}^{M}V_\vartheta K_\vartheta(t),\qquad B(t)=\sum_{\vartheta=0}^{M}B_\vartheta K_\vartheta(t),\qquad Z(t)=\sum_{\vartheta=0}^{M}Z_\vartheta K_\vartheta(t),\end{aligned}\tag{19}$$

where$S_\vartheta, L_\vartheta, I_\vartheta, V_\vartheta, B_\vartheta, Z_\vartheta$ are the Legendre coefficients corresponding to each class. Implementing the provided estimate, we can simplify Eq. 17 into the following form, making them more convenient to apply:

$$\sum_{\vartheta=0}^{M}S_nK_\vartheta(t)=\sum_{\vartheta=0}^{M}S_nK_\vartheta(0)+\frac{t}{2}\sum_{k=0}^{M}\{\lambda-d\sum_{\vartheta=0}^{M}S_nK_\vartheta(t)$$

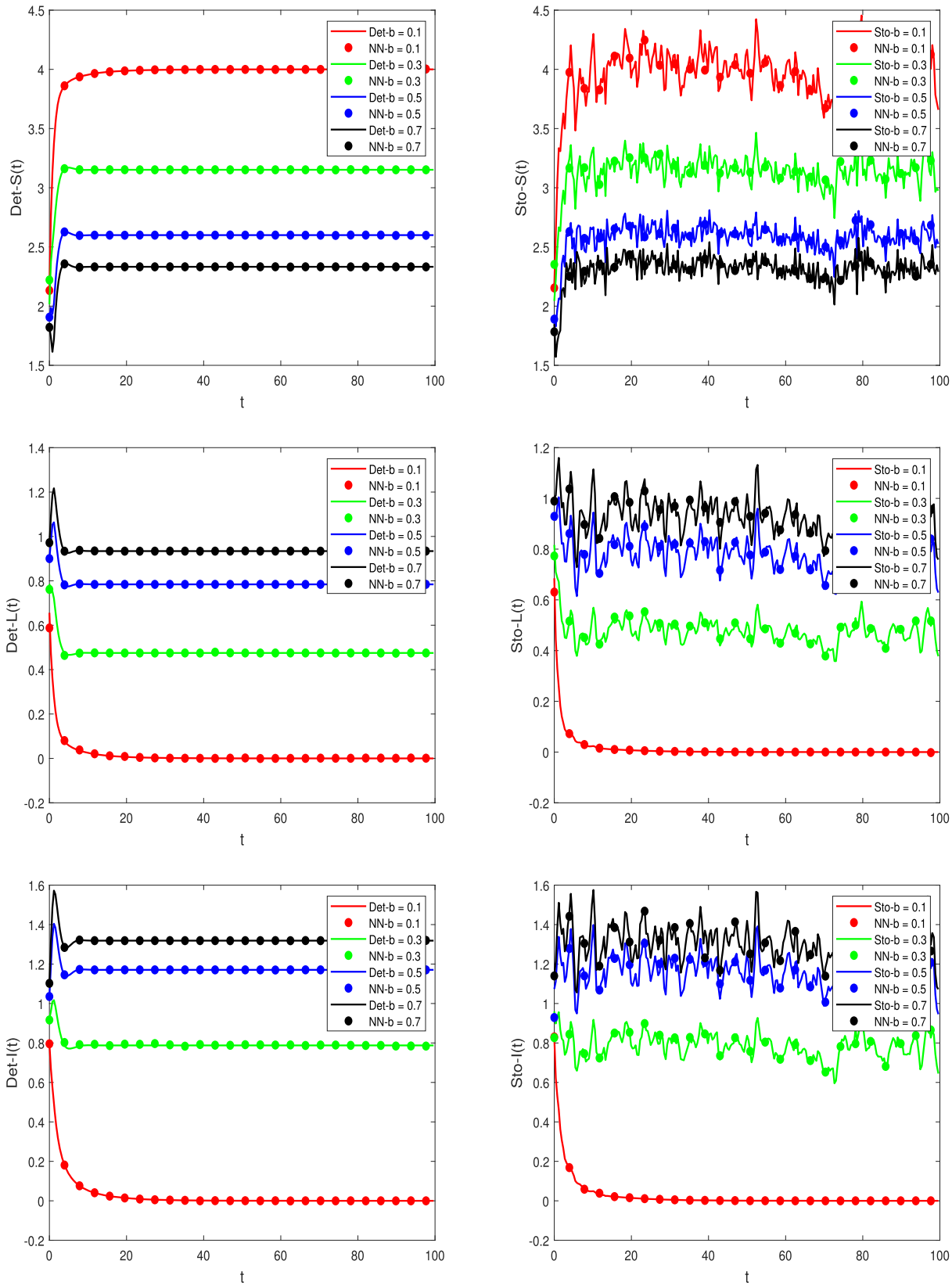


**Fig. 17.** Simulation illustrating the effect of $b$ as $b = 0.2, 0.4, 0.6$ and 0.8 on the dynamics of Susceptible Cells, Latently Infected Cells, and Infected Cells. The plot shows the comparison between deterministic and stochastic models along with FFNN approximation.

$$- b\sum_{\vartheta=0}^{M} S_n K_\vartheta(t)\sum_{\vartheta=0}^{M} V_n K_\vartheta(t)\big\}W_k + \frac{t}{2}\sum_{k=0}^{M}\big\{\sigma_1\sum_{\vartheta=0}^{M} S_n K_\vartheta(t)dW_1 \times \frac{t}{2}(v_0+1)\big\}W_k^*,$$

$$\sum_{\vartheta=0}^{M} L_n K_\vartheta(t) = \sum_{\vartheta=0}^{M} L_n K_\vartheta(0) + \frac{t}{2}\sum_{k=0}^{M}\big\{(1-\rho)b\sum_{\vartheta=0}^{M} S_n K_\vartheta(t)\sum_{\vartheta=0}^{M} V_n K_\vartheta(t) - (\theta+\pi)\sum_{\vartheta=0}^{M} L_n K_\vartheta(t)\big\}W_k + \frac{t}{2}\sum_{k=0}^{M}\big\{\sigma_2\sum_{\vartheta=0}^{M} L_n K_\vartheta(t)dW_2 \times \frac{t}{2}(v_0+1)\big\}W_k^*,$$

$$\sum_{\vartheta=0}^{M} I_n K_\vartheta(t) = \sum_{\vartheta=0}^{M} I_n K_\vartheta(0) + \frac{t}{2}\sum_{k=0}^{M}\big\{\rho b\sum_{\vartheta=0}^{M} S_n K_\vartheta(t)\sum_{\vartheta=0}^{M} V_n K_\vartheta(t) + \pi \times \sum_{\vartheta=0}^{M} L_n K_\vartheta(t) - a\sum_{\vartheta=0}^{M} I_n K_\vartheta(t) - \epsilon\sum_{\vartheta=0}^{M} I_n K_\vartheta(t) Z(\frac{t}{2}(v_0+1))dr\big\} W_k + \frac{t}{2}\sum_{k=0}^{M}\big\{\sigma_3\sum_{\vartheta=0}^{M} I_n K_\vartheta(t)dW_3\frac{t}{2}(v_0+1)\big\}W_k^*,$$

$$\sum_{\vartheta=0}^{M} V_n K_\vartheta(t) = \sum_{\vartheta=0}^{M} V_n K_\vartheta(0) + \frac{t}{2}\sum_{k=0}^{M}\big\{m\sum_{\vartheta=0}^{M} I_n K_\vartheta(t) - r\sum_{\vartheta=0}^{M} V_n K_\vartheta(t) - q \times \sum_{\vartheta=0}^{M} V_n K_\vartheta(t)\sum_{\vartheta=0}^{M} B_n K_\vartheta(t)dr\big\}W_k + \frac{t}{2}\sum_{k=0}^{M}\big\{\sigma_4\sum_{\vartheta=0}^{M} V_n K_\vartheta(t)dW_4 \frac{t}{2}(v_0+1)\big\}W_k^*,$$

$$\sum_{\vartheta=0}^{M} B_n K_\vartheta(t) = \sum_{\vartheta=0}^{M} B_n K_\vartheta(0) + \frac{t}{2}\sum_{k=0}^{M}\big\{\eta + c\sum_{\vartheta=0}^{M} B_n K_\vartheta(t)\sum_{\vartheta=0}^{M} V_n K_\vartheta(t) - \sigma\sum_{\vartheta=0}^{M} B_n K_\vartheta(t)dr\big\}W_k + \frac{t}{2}\sum_{k=0}^{M}\big\{\sigma_5\sum_{\vartheta=0}^{M} B_n K_\vartheta(t)dW_5\frac{t}{2}(v_0+1)\big\}W_k^*,$$

$$\sum_{\vartheta=0}^{M} Z_n K_\vartheta(t) = \sum_{\vartheta=0}^{M} Z_n K_\vartheta(0) + \frac{t}{2}\sum_{k=0}^{N}\big\{\gamma + w\sum_{\vartheta=0}^{M} I_n K_\vartheta(t)\sum_{\vartheta=0}^{M} Z_n K_\vartheta(t) - \mu\sum_{\vartheta=0}^{M} Z_n K_\vartheta(t)dr\big\}W_k + \frac{t}{2}\sum_{k=0}^{N}\big\{\sigma_6\sum_{\vartheta=0}^{M} Z_n K_\vartheta \times (t)dW_6\frac{t}{2}(v_0+1)\big\}W_k^*. \tag{20}$$

Taking $\eta_k = \left(\frac{t}{2}(v_0+1)\right)$, then there are $6M+6$ unknowns with $6M$ nonlinear algebraic equation in the system (20) as follows:

$$\sum_{\vartheta=0}^{M} S_n K_\vartheta(0) = \alpha_1, \quad \sum_{\vartheta=0}^{M} L_n K_\vartheta(0) = \alpha_2, \quad \sum_{\vartheta=0}^{M} I_n K_\vartheta(0) = \alpha_3,$$
$$\sum_{\vartheta=0}^{M} V_n K_\vartheta(0) = \alpha_4, \quad \sum_{\vartheta=0}^{M} B_n K_\vartheta(0) = \alpha_5, \quad \sum_{\vartheta=0}^{M} Z_n K_\vartheta(0) = \alpha_6. \tag{21}$$

As a result, Eqs. 20 and 21 constitute a nonlinear algebraic system comprising $(6M+6)$ unknowns, corresponding to $S_\vartheta, L_\vartheta, I_\vartheta, V_\vartheta, B_\vartheta, Z_\vartheta$ for $\vartheta = 0, 1, \ldots, M$. By substituting these unknowns into system (2), we can derive an iterative solution for the model that accounts for stochastic effects.

### 4.2. Feedforward neural network scheme

A FFNN is a computational model inspired by the structure of the biological neural system. It consists of one input and output layer each and three hidden layers. Each layer performs a linear transformation followed by a nonlinear activation function. This architecture enables the modeling of complex nonlinear relationships in the data and improves the accuracy of solution approximation.

Let $\mathbf{x} \in \mathbb{R}^d$ denote the input feature vector for the neural network, where $d$ is the number of collocation points in the spectral method. Each component $x_i$ of $\mathbf{x}$ corresponds to the approximate value of the solution at the $i$th collocation point, obtained via the spectral collocation technique. This input enables the neural network to learn the mapping between the numerical solution space and the desired output behavior. The FFNN maps $\mathbf{x}$ to an output $\hat{y} \in \mathbb{R}$ through the following sequence of transformations:

$$\mathbf{z}^{(1)} = \mathrm{lmb}\big(\mathcal{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}\big),$$
$$\mathbf{z}^{(2)} = \mathrm{lmb}\big(\mathcal{W}^{(2)}\mathbf{z}^{(1)} + \mathbf{b}^{(2)}\big),$$
$$\mathbf{z}^{(3)} = \mathrm{lmb}\big(\mathcal{W}^{(3)}\mathbf{z}^{(2)} + \mathbf{b}^{(3)}\big),$$
$$\hat{y} = \mathbf{W}^{(4)}\mathbf{z}^{(3)} + \mathbf{b}^{(4)},$$

where

- $\mathcal{W}^{(i)}$ and $\mathbf{b}^{(i)}$ are the weight matrices and bias vectors for layer $i$,
- $\mathbf{z}^{(i)}$ represents the output (activation) from layer $i$,
- $\mathrm{lmb}(\cdot) = \frac{1}{1+e^{-x}}$ is the log-sigmoid activation function.

The set of trainable parameters are denoted by:

$$\Theta = \left\{\mathcal{W}^{(i)}, \mathbf{b}^{(i)}\right\}_{i=1}^{4}.$$

#### 4.2.1. Nonlinear activation functions

Each hidden layer applies a nonlinear activation function $\beta(\cdot)$ to introduce nonlinearity.

- **Log-Sigmoid :** $\beta(z) = \frac{1}{1+e^{-z}}$

We employ the **log-sigmoid** activation function for all hidden layers, as it provides smooth output in the range $(0, 1)$ and is well-suited for nonlinear problems.

#### 4.2.2. Loss function and error computation

To train the network, a loss function $\mathcal{L}(\Theta)$ is defined to measure the discrepancy between the predicted output and the actual target values. For our regression-based problem, we employ the MSE, which is given by:

$$\mathcal{L}(\Theta) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2, \tag{22}$$

where

- $n$ is the number of training samples,
- $y_i$ is the true (target) output corresponding to the $ith$ sample,
- $\hat{y}_i$ is the predicted output from the neural network,
- $\Theta$ denotes the set of all trainable parameters, including weights and biases.

This loss function is minimized during training to improve the network prediction accuracy.

#### 4.2.3. Training algorithm

The network parameters $\Theta$ are optimized to minimize the loss function using the Levenberg Marquardt (LM) algorithm. This method is particularly effective for small to medium-sized feedforward neural networks and is well-suited for nonlinear least squares problems. The LM update rule is defined as:

$$\Theta^{(k+1)} = \Theta^{(k)} - \left[J^\top J + \lambda I\right]^{-1} J^\top \mathbf{e}, \tag{23}$$

where

- $\Theta^{(k)}$ is the parameter vector at iteration $k$,
- $J$ is the Jacobian matrix of partial derivatives of the network errors with respect to $\Theta$,
- $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ is the error vector,

- $\lambda$ is a non-negative damping factor that controls the step size,
- $I$ is the identity matrix.

The LM algorithm combines the advantages of both Gradient Descent and the Gauss Newton method:

- When $\lambda$ is large, the update behaves like Gradient Descent:
$$\Theta^{(k+1)} \approx \Theta^{(k)} - \frac{1}{\lambda} J^\top \mathbf{e}.$$
- When $\lambda$ is small, it behaves like the Gauss Newton method:
$$\Theta^{(k+1)} \approx \Theta^{(k)} - \left(J^\top J\right)^{-1} J^\top \mathbf{e}.$$

During training, $\lambda$ is dynamically adjusted: it decreases when the loss reduces (favoring Gauss Newton steps) and increases when the loss increases (falling back to Gradient Descent for stability). This adaptive behavior ensures both convergence speed and robustness. In our implementation, MATLAB's `trainlm` function is used to apply this algorithm. The procedure for the numerical simulation is obtain using the Algorithm 1.

**Algorithm 1** A Hybrid algorithm based on LSCM-FFNN for solving within-host CHIKV stochastic transmission model.

**Input:** Initial values i.e., $S(0), L(0), I(0), V(0), B(0), Z(0)$, time domain $[0, T]$, number of collocation points $N$
**Output:** Approximate solution for each class in the model over the defined time interval $[0, T]$
**Step 1: Define SLIVBZ Model**
For each compartment Define a nonlinear stochastic differential equations
**Step 2: Apply Legendre Spectral Collocation Method**
**for** each compartment $X(t) \in \{S, L, I, V, B, Z\}$ **do**
    Approximate with Legendre polynomials $P_k$ as $X(t) \approx \sum_{k=0}^{N} c_k P_k(\xi)$
    Evaluate at Gauss-Lobatto collocation points to construct an algebraic system
**end for**
**Step 3: Design and Train FFNN**
Construct a FFNN with input $X(t) \in \mathbb{R}^6$, which produce approximations of nonlinear terms as an output
Train the FFNN using supervised learning to minimize prediction error for model nonlinearities
**Step 4: Hybrid Integration**
**for** each time-step $t_n$ from 0 to $T$ **do**
    Use FFNN to compute $\hat{f}(X(t_n))$, an approximation of the nonlinear systems
    Update each class using:
$$X(t_{n+1}) = X(t_n) + \Delta t \cdot \hat{f}(X(t_n)) + \Delta W(t_n)$$
    where $\Delta W(t_n)$ denotes the stochastic noise term
**end for**
**Step 5: Post-processing**
Analyze performance of the scheme
Depicts dynamics of $S, L, I, V, B, Z$

## 5. Simulation and discussion

To validate our theoretical results, we present comprehensive simulation for both the stochastic and deterministic systems presented in (1) and (2) respectively. The numerical findings confirm the robustness of the CHIKV model and support the theoretical analysis, demonstrating the reliability of the scheme. Additionally, the numerical simulation offers a practical insight into implementing the suggested solutions. The parameter values used in the simulation are obtained from the deterministic model in existing literature [19], while a distinct parameter set is utilized to examine the global behavior of the stochastic model at equilibrium. Fig. 1

In the first case we consider the set of parameters as: $\lambda = 4$, $\mu = 1.20$, $c = 1.21290$, $d = 0.7979$, $\gamma = 0.50$, $w = 0.50$, $\epsilon = 0.4441$, $\sigma = 0$, $\rho = 0.50$, $\theta = 0.50$, $\pi = 0.5$, $m = 2.020$, $r = 0.442$, $q = 0.596$, $\eta = 1.4020$, $a = 0.50$, $\delta = 1.2510$, $\alpha = 0.50$, $b = 0.10$. The initial values considered in this case are $S_0 = 1.9$, $L_0 = 1.4$, $I_0 = 0.9$, $V_0 = 1$, $B_0 = 1$, $Z_0 = 1.2$. The step size is considered as $\Delta t = \frac{1}{256}$. Using these parameter values, the basic reproduction number of the deterministic model remains less than one, i.e., $\mathcal{R}_0 < 1$. The resulting graphical results are depicted in Fig. 2. These plots clearly illustrates that the deterministic model (1) reaches a stable CHIKV free equilibrium, suggesting the eradication of the infection over time. This result aligns with the theoretical expectation that when $\mathcal{R}_0 < 1$, the infection cannot sustain itself within the population and would ultimately dies out. The initial values considered in this case are The simulation results for the stochastic model in this case and with $\bar{\mathcal{R}}_0 < 1$ are presented in Fig. 3. Based on these findings, it can be concluded that the CHIKV free state $\mathcal{E}_0$ remains stable under the given stochastic conditions, indicating that the infection will not persist within the population. Fig. 4 presents a comparison simulation of the stochastic and deterministic within-host CHIKV models as described in (1) and (2) respectively. Furthermore, we consider $\sigma_i = 0.5$ where $i = 1, 2, \ldots, 6$. Consequently, Fig. 4 clearly illustrates that both the deterministic and stochastic systems satisfy the criteria for CHIKV-free equilibrium, ensuring the extinction of the virus.

In the second case, we consider the same set of parameters as in the first case, except for $b = 0.90$, and the resulting simulations are depicted in Figs. 5–7. With these parameter values and $\mathcal{R}_0 > 1$, along with $\sigma_i = 0.5$ where $i = 1, 2, \ldots, 6$, Fig. 5 confirms the stability of EE. Fig. 6 further validates that the stochastic case model (2), accurately captures the transmission dynamics of CHIKV. The simulation obtained with this set of parameters demonstrate that the value of $\bar{\mathcal{R}}_0$ exceeds 1, indicating that the infection persists in the population. Additionally, the model converges to a stable EE denoted as $\mathcal{E}^*$. A comparison of both the cases CHIKV epidemic models, as formulated in Eqs. (1) and (2), is conducted in Fig. 7. For each class, the comparison between the deterministic and stochastic systems reveals that both models exhibit a stable EE, represented as $\mathcal{E}^*$. The simulation results conclusively indicate that CHIKV will persist in the population.

Figs. 8 and 9 illustrate the impact of the standard Wiener process intensity, $\sigma_i$ (where $i = 1, 2, \ldots, 6$), on each compartment of the stochastic system described in Eq. (2). The simulations are conducted for varying values of $\sigma_i = 0.1, 0.5, 0.9$, demonstrating the influence of stochastic fluctuations on CHIKV dynamics.

The neural networks architecture used in the present study is depicted in Fig. 10. The graphical procedure of the present study is analyzed in Fig. 11. It has three hidden layers, each with 50 nodes, after a single input layer. The output layer consists of six nodes, corresponding to main classes of the model. This approach aims to achieve precise predictions and computational efficiency while effectively capturing the intricate dynamics of CHIKV.

Fig. 12 presents a comparative analysis of the solutions obtained from the deterministic model and those generated using the NN approach, alongside a comparison between the stochastic model solutions and the NN approach for the case $R_0 < 1$. These plots demonstrate that the solutions produced by both the conventional method and the NN approach exhibit excellent agreement. Fig. 13 presents the optimal validation results, including regression analysis, error histograms, and gradient evaluations, utilized to analyze the CHIKV within host system for $R_0 < 1$, as shown in Fig. 12. These findings highlight the accuracy of the NN approach, demonstrating minimal error between the predicted and actual solutions, as well as the effectiveness of the models training and validation processes. Additionally, the gradient results confirm the optimization efficiency throughout the training phase.

Fig. 14 compares the solutions of the CHIKV within host deterministic system with those forecasted by the FFNN alongside the solutions

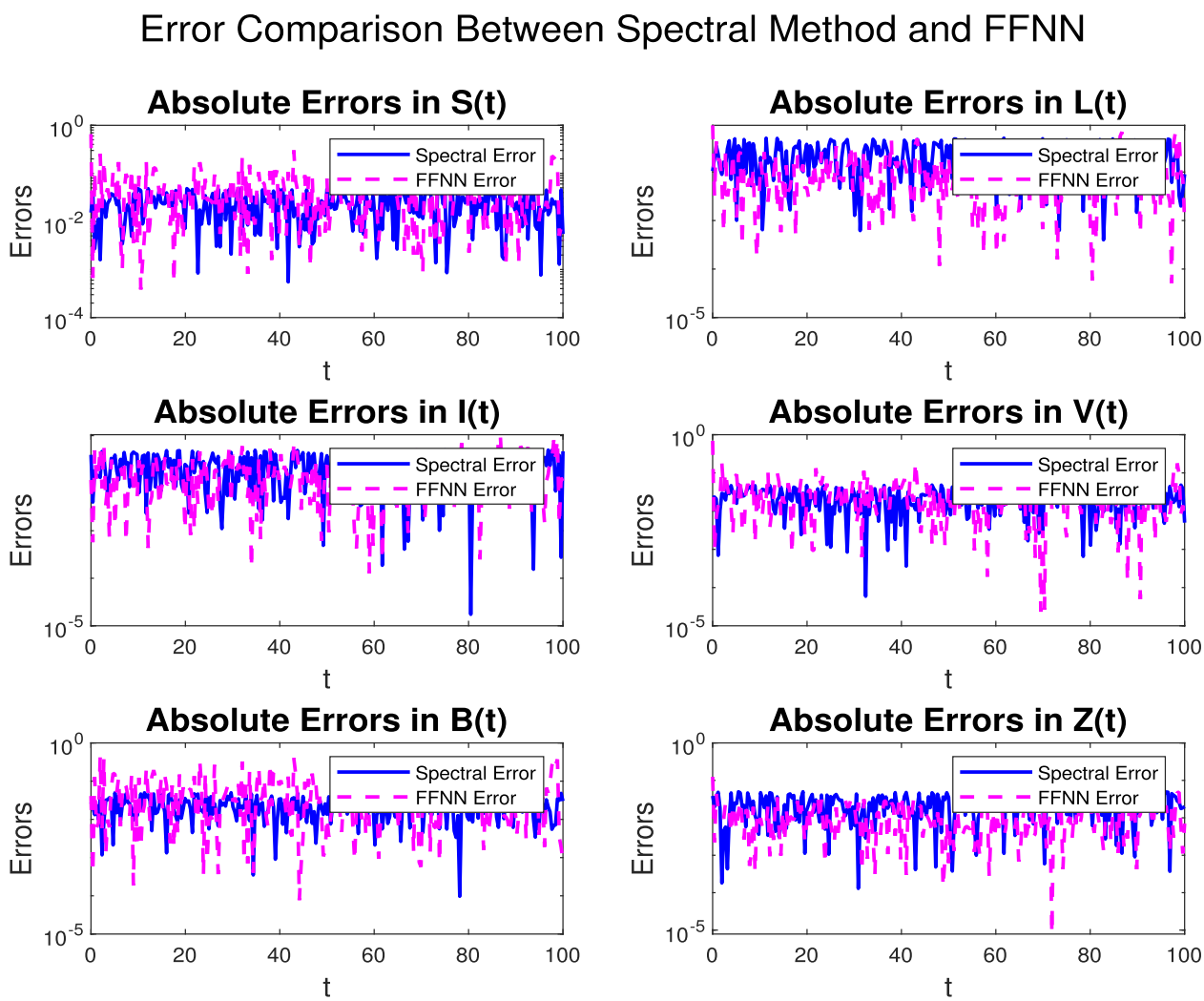


**Fig. 18.** Error comparison between Spectral method and FFNN.

of the stochastic system and the FFNN predictions, for $R_0 > 1$. The parameter values used are consistent with those mentioned in the previous figures.

Fig. 15 presents the best validation results, including regression analysis, error histograms, and gradient outcomes, for solving the CHIKV model when $R_0 > 1$, as illustrated in Fig. 14. The neural network performance is illustrated by these results, which emphasize the effectiveness of the training and validation procedures, minimal error between predicted and actual solutions, and accuracy. The gradient results further validate the optimization efficiency during the training phase.

Fig. 16 illustrates the behavior of each class under varying values of $\rho$. We consider four distinct values, $\rho = 0.1, 0.3, 0.5$, and 0.7, to analyze the impact of this parameter (latency rate) while keeping all other parameters consistent with those used in the previous plots. The graph highlights how changes in $\rho$ influence the dynamics of each class within the CHIKV epidemic model, offering insights into the system sensitivity to fluctuations in the latency rate.

In the last Fig. 17, the dynamics of each state variable in the model are assessed under different transmission rates $b$. The specific values considered in the simulation are $b = 0.2, 0.4, 0.6$, and 0.8, while keeping all other parameters consistent with those used in the preceding plots. The resulting plots in Fig. 17 illustrate the impact of variations in the transmission rate on the dynamics of each class within the CHIKV epidemic model, offering insights into the system sensitivity to fluctuations in $b$.

## 6. Neural network based simulation

Recently, neural network based simulation has become essential for epidemic prediction, offering significant advantages over traditional computational methods. This study explores the computational dynamics of the stochastic model (2) using the FFNN approach outlined in [39]. The training dataset consists of input values corresponding to mesh nodes distributed over the interval, which determine the system output, referred to as the solution. These nodes encapsulate key components of the system functionality. The ODEs are numerically solved using the odeint solver from the SciPy library, generating a dataset that accurately represents the system progression. This dataset serves as a robust foundation for training and analysis. While the deterministic case is solved using the odeint solver, the stochastic case is handled using the LSCM. An effective approach to enhancing NN performance is normalizing the training data within the range of 0 to 1. This can be achieved by adjusting the system architecture and replacing component densities in the updated model. The model output represents the rate of change of the dependent variable, expressed as a function of the system parameters and state variables. Additionally, we employ the linspace function from the NumPy package, where $t_{\text{initial}}$ defines the starting time, $t_{\text{final}}$ specifies the endpoint, and the chosen number of subdivisions determines the step size. This function generates an array that establishes the integration interval for the system under consideration. $X_{\text{train}}$ represents the training set. To enhance the accuracy of the training process, we increase

the number of subdivisions, thereby incorporating additional independent variables. Selecting an appropriate NN architecture is crucial for epidemiological simulations, as it directly influences model efficiency, data learning precision, and predictive performance. The NN architecture consists of various structural components, including the number of layers, the number of neurons per layer, and the connectivity patterns between them. A key aspect of architectural design is optimizing computational efficiency while minimizing processing demands. The model must be sufficiently adaptable to accommodate evolving disease dynamics and varying data volumes. Striking this balance ensures both accurate epidemic representation and computational efficiency.

A neural network is constructed using the feedforward net function, and training is performed using the Levenberg Marquardt algorithm via the trainlm function in MATLAB. The FFNN procedure is structured with three hidden layers, each comprising 50 neurons. During the training process, the network is set to run for 1000 epochs. Plotting the loss decay across the training period will help one observe the model development. Initially, the loss appears to be low, indicating a fast learning and improvement mechanism of the model. As the number of epochs increases, the rate of enhancement decreases, and the curve representing the loss function begins to stabilize, suggesting that the system is approaching optimal performance.

The preparation phase begins with defining the ODE system, specifying the initial conditions, and importing the necessary libraries. The stochastic ODE problem is solved using the Legendre spectral collocation method during the data preparation phase by discretizing the obtained solutions into training grids. Furthermore, an FFNN based problem is formulated with appropriate inputs, hidden layers, and output layers. The model is then constructed using the Levenberg Marquardt technique, with the mean squared error serving as the loss function. By comparing the system predictions with the spectral solution over the same interval, the model can be effectively trained and evaluated. As a powerful tool for understanding and solving complex differential equations, the trained model is applied to generate predictions and perform analysis.

After 43 epochs, the best performance values for various neural network components training, validation, and testing are recorded as $-3.8 \times 10^{-4}$, $-1.200 \times 10^{-3}$, and $-2.50 \times 10^{-4}$, respectively. Additionally, at 14 epochs, the gradient value is $1.1517 \times 10^{-4}$ for the deterministic approach when $R_0 < 1$. For the case where $R_0 > 1$, the deterministic case model achieves its results after 61 epochs, with performance values of $5.5 \times 10^{7}$, $-3.2 \times 10^{-4}$, and $6.20 \times 10^{-4}$, respectively. The gradient value at epoch 61 is recorded as $1.32 \times 10^{-4}$. When analyzing the system dynamics under different conditions of $R_0$, the obtained results confirm the convergence, accuracy, and precision of the developed ANN. The computation times using ANN range from 23 to 27 minutes for certain scenarios and from 32 to 36 minutes for others, highlighting the efficiency of these mathematical models in solving equations for both deterministic and stochastic systems.

### 6.1. Absolute error comparison of FFNN and LSCM

In this section, we evaluate the accuracy and computational performance of the FFNN-based approach in comparison to the conventional LSCM, applied across all compartments of the stochastic CHIKV within-host model. The comparison of both techniques is presented in terms of absolute errors over time. Fig. 18 presents the absolute errors of the LSCM and FFNN methods for solving the CHIKV model. This comparison highlights the efficacy and accuracy of each method by demonstrating how well they approximate the system's solutions. The effectiveness and precision of the spectral collocation technique in capturing the dynamics of the CHIKV model can be assessed by comparing its error metrics with those of the FFNN approach. This evaluation provides crucial insights into the advantages and limitations of each method, aiding in the selection of the most suitable approach for various applications in epidemiological modeling.

## 7. Conclusion

In this paper, we introduced an efficient hybrid numerical computing approach for analyzing the complex within-cell dynamics of CHIKV under both deterministic and stochastic environments. Stochastic computing with feedforward neural networks is integrated with the Levenberg-Marquardt backpropagation method to investigate the persistence and spread of CHIKV under randomness. The well-established Legendre spectral scheme is employed as a reference tool or target dataset for neural computing. The threshold parameters $R_0$ and $\bar{R}_0$ are established, and it is numerically demonstrated that, when the control reproduction number is less than unity, the infected cell count of CHIKV diminishes, leading to the elimination of the virus. Conversely, when the control reproduction number exceeds unity, a unique stable endemic equilibrium is established, ensuring the persistence of the infection. Furthermore, the dynamics of latent and actively infected cells are analyzed under varying latency and transmission rates. The impact of the standard Wiener process intensity is illustrated graphically, demonstrating the influence of stochastic fluctuations on CHIKV dynamics. Furthermore, by randomly selecting training, testing, and validation samples from the reference dataset, the NN-based model produces an approximate solution for the stochastic nonlinear CHIKV model, exhibiting high consistency with the benchmark simulation. Finally, simulation outcomes, including error distribution histograms, least mean squared errors, and regression evaluations, further validate the proposed ANNs' effectiveness, reliability, and accuracy in solving the stochastic nonlinear CHIKV within-host model across different scenarios. We believe that the robust neural network paradigm presented in this study also carries significant practical implications. By identifying threshold conditions for viral eradication or persistence, the model provides insights into treatment strategies aimed at reducing viral replication below epidemic thresholds. In future work, we intend to extend the proposed technique to more complex epidemic control problems to achieve improved strategies for minimizing infection.

## CRediT authorship contribution statement

**Shuo Li:** Writing – review & editing, Supervision, Software, Methodology, Conceptualization; **Misbah Ullah:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Saif Ullah:** Validation, Supervision, Software, Methodology, Conceptualization; **Taseer Muhammad:** Writing – review & editing, Validation, Supervision, Software, Project administration, Investigation; **Qaiser Iqbal:** Writing – review & editing, Visualization, Software, Formal analysis.

## Data availability

No data was used for the research described in the article.

## Declaration of competing interest

The authors declare no conflict of interest.

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